28. What protects the brain?
A. Membrane and neck bone
B. Membrane and back bone
C. Membrane and skull bone
D. Membrane and rib bone

Explanation: The brain is protected by:
- ✓ Skull
- ✓ Membrane (meniges)
- ✓ Cerebra spinal fluid

Answer: C

Post Test – 2

I. Choose the best option from the given alternatives.

1. Which of the following are parts of the central nervous system?
A. Spinal nerve and cranial nerve
B. Spinal nerve and spinal cord
C. Spinal cord and brain
D. Brain and cranial nerve

The following items are based on the list below.

i. Sensory receptor
ii. Afferent neuron
iii. Response
iv. Efferent neuron
v. stimulus
vi. Central nervous system
vii. Effector organ

2. The correct path of an impulse during nervous coordination is:
A. i → v → vii → vi → iv → iii → ii
B. v → i → ii → vi → iv → vii → iii
C. i → ii → iii → iv → v → vi → vii
D. iv → iii → vii → v → vi → i v ii

3. organs of the body such as muscles and glands are:
A. v B. iii C. vii D. iv

4. The part of our nervous system that detects a stimulus is:
A. i B. ii C. iv D. vii

5. All of the following are the protective structures of the brain except:
A. Vertebral column
B. Cerebrospinal fluid
C. Cranium
D. Meninges

6. Nerves that are directly connected to the brain are:
A. Cranial nerves
B. Spinal nerves
C. Sensory nerves
D. Motor nerves

7. Gray matter of the brain consists of:
   A. Cell bodies and axon of the neuron.
   B. Cell bodies of neurons and synapse
   C. Dendrite of neurons and synapse
   D. Axon and dendrites of a neuron
8. The part of the brain that is concerned in coordination such as thinking, speech, memory, intelligence is:
   A. Cerebellum
   B. Medulla oblongata
   C. Cerebrum
   D. Hypothalamus
9. All of the following are the functions of hypothalamus except:
   A. Regulation of body temperature
   B. Feeling of hunger and thirst
   C. Fat metabolism
   D. Balance and posture
10. The vision center in the brain is:
   A. Cerebellum
   B. Mid brain
   C. Cerebrum
   D. Thalamus
11. Involuntary movements of the body are controlled by:
   A. Medulla oblongata
   B. Cerebellum
   C. Cerebrum
   D. Optic lobe
12. Vital nerve center for controlling heartbeat, respiration, blood pressure and breathing is:
   A. Cerebrum
   B. Cerebellum
   C. Medulla oblongata
   D. Hypothalamus

Voluntary and Refax control

The response of our body to external environment can be classified as **voluntary** and **reflex control.**

- ✓ **Voluntary control** – are activities of the body that we choose to do. For instance when you see a mango or banana that is ripe on a market; you may pick it up, feel and smell it before deciding if it is as good as it looks and buying it.
- ✓ **Reflex action** is a sudden automatic and uncontrolled response of parts of the body or the whole body to external stimuli.

Step 3: Ringing of bell every time the foodwas given. This was repeated number of times. The dog salivates as a food is presented after the ringing of bell. The dog started to associate the ringing of bell with the presence of food, because the ringing of bell became a conditioned reflex.Other examples of reflex action include:

- Type writing, playing key board, etc.

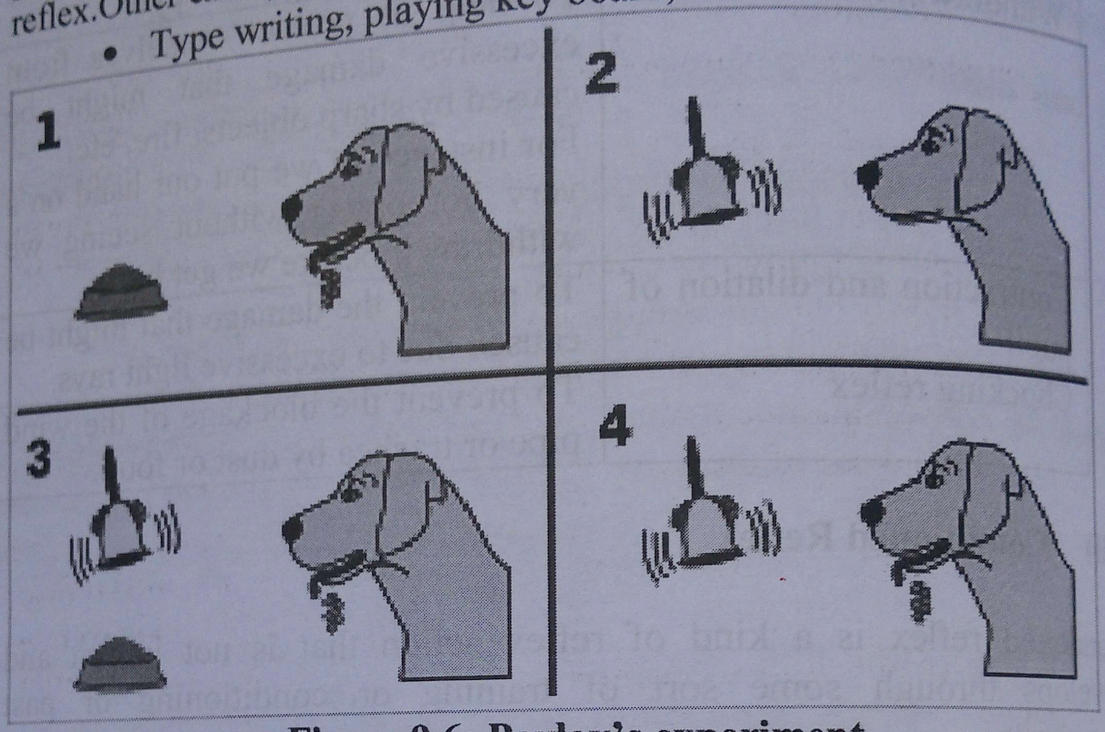


Figure 9.6: Pavlov's experiment

Illustrative Example

29. In a simple reflex, the path way of a neuron is:
    A. Sensory neuron → motor neuron →association neuron
    B. Motor neuron → association neuron → sensory neuron
    C. Association neuron → sensory neuron → motor neuron
    D. Sensory neuron → association neuron → motor neuron

    Explanation: Reflex arc is a pathway of neuron during reflex action. The pathway of an impulse during a reflex action is Receptors → sensory neuron → association neuron → motor neuron → effectors **Answer: D**

30. Neuron in a spinal reflex make synapses in the
    A. Dorsal root
    B. Ventral root
    C. Grey matter
    D. White matter

    Explanation: The afferent and efferent neurons meet with relay neuron in the grey matter of the spinal cord. **Answer: C**

31. An example of conditioned reflex is:
    A. Type writing
    B. Swallowing
    C. Knee jerk
    D. Sneezing

Explanation: Conditioned reflex is not in born and develops after some sort of conditioning or training past experience.

Answer: D

32. In reflex action, to which of the following does the impulse reach last?
A. Coordinator
B. Receptors
C. Efferent neuron
D. Effector

Explanation: The pathway of an impulse during reflex action is: Stimulus → receptor → afferent neuron → coordinator → afferent neuron → effector organ

Answer: D

33. Which of the following is conditioned reflex?
A. Salivating on seeing food.
B. Salivating on tasting food.
C. Sneezing doe to dust
D. Blinking of the eye

Explanation:
Salivating up on seeing or smelling a food is a conditioned reflex.

Answer: A

34. Which of the following situations is likely to produce a reflex action?
A. Seeing a meal
B. Receiving a present
C. Climbing a tree
D. Reading a text book

Explanation: Seeing a meal may make a person to salivate. This is a conditioned reflex.

Answer: A

35. The major difference between reflex action and normal conscious action is that, in flex action?
A. The response is very slow.
B. Relay neurons play the role of coordination.
C. The coordinator is the spinal cord or the brain.
D. Information does not reach the brain or spinal cord.

Explanation: Reflex action is a rapid automatic response to a stimuli. It is mainly coordinated by the spinal cord.

Answer: C

36. Which of the following is true about reflex action? It is
A. Slow response to the stimuli
B. Fast response to the stimuli
C. Learned response to stimuli
D. Controlled response to stimuli

Explanation: Reflex action is
- ✓ Fast or automatic response to stimuli
- ✓ Uncontrolled and in born

Answer: B

37. This item is based on the following table which compares reflex action and conditioned reflex. Which one

No.	Reflex action	Conditioned reflex
A	It is a result of experience	It is innate behavior
B	Only brain act as a coordinator	Only spinal cord act as coordinator
C	It is an automatic and uncontrolled process	It is an experience and learning process
D	Moving away from ant by a child is an example	Breathing is an example

Explanation:
The correct comparison is as follows:

Reflex action	Conditioned reflex
✓ It is innate (in born) behavior. ✓ Spinal cord acts as a coordinator ✓ It is automatic and unconditioned	✓ It is a result of experience learning ✓ Coordinating the brain

Answer: C

38. Which of the following statement is correct about the reflex arc? It is
- A. Flow or impulse from sensory neuron to the brain and spinal cord.
- B. Flow of impulse from the spinal cord to the brain and back to spinal cord.
- C. Flow of information from receptors to the spinal cord and to the effectors.
- D. Unidirectional flow of information from receptros to spinal cord.

Explanation:
Reflex arc is the flow of an ampules during a reflex action. It involves the flow of information from the receptors to spinal cord and to the effectors.

Answer: C

Post Test – 3

Choose the best option from the given alternatives.

1. The pathway of an impulse during reflex action is:
   A. Coordination
   B. Simple reflex
   C. Conditioned reflex
   D. Reflex arc

2. During reflex action impulse enters the spinal cord through:
   A. Ventral root
   B. Spinal nerve
   C. Dorsal root
   D. Vertebral column

3. Which of the following is NOT a reflex action?
   A. Blinking
   B. Eating
   C. Chocking
   D. Sneezing

4. A reflex action that is used to prevent the blockage of the wind pipe by dust or food is:
   A. Blinking
   B. Withdrawal
   C. Chocking
   D. Knee jerk

5. Reflex action is mainly coordinated by:
   A. Brain
   B. Spinal cord
   C. Effectors
   D. Receptors

6. Simple reflex action differs from conditioned reflex by:
   A. Simple reflex is learned
   B. Conditioned reflex is unlearned
   C. Simple reflex is inborn
   D. Conditioned reflex is without prior experience

7. In Ivan Pavlov's experiment on dog, which of the following is a conditioned reflex?
   A. Salivation of the dog
   B. Ringing of the bell
   C. Appearance of the food
   D. Hunger by the dog

Drug Abuse

What is drug?

Drug refers to any kind of chemical substance that can affect the working of the body when taken or used. People use drug for several Porposes, Among these, some of them are:

- ✓ For recreational purpose
- ✓ For coping with anxious or stressful situations
- ✓ For medical purpose, etc

Drugs have a tendency to alter/change the way wind, body or both work. Drugs can be classified**legal** or **illegal** based on their impact and necessity by the body.

Illegal drugs

Illegal drugs are drugs that do not have any medicinal significance and affect the physiological and psychological state of the user.
Illegal drugs are banned by nations and it is a crime to smuggle, carry, sell and use them. The type of drug that is considered illegal varies from one country to another. However there are some drugs that are taken as illegal in most countries. These are:

- ✓ Cannabis
- ✓ Cocaine
- ✓ Marijuana
- ✓ Lysergic acid, etc

Legal drugs

Legal drugs are drugs that are used for medicinal purpose. Some drugs that do not have adverse effect and used for recreational purpose are also considered as legal.
Legality of the drug also depends on their social value in the society. Such drugs are known as socially accepted recreational drugs. For instance in Ethiopia people use mild stimulant drugs such as **caffeine (found in tea, coffe, coca cola), Khat (ጫት)** and**traditional alcoholic drinks** (tej, tella, etc).
Even if there is no legal ground that prevents the use of drugs such as khat and alcohol, it does not mean that they have adverse health effects.

For medicinal drugs, it is advisable and appropriate to use only a **prescribed drug**. It is also import to take appropriate dosage and look at its expiry date. Unless they may have adverse effects on health.

Prescribed drug is a drug that is aided or recommended by a physician after proper diagnosis.

Point of discussion
Make a discussion with your classmates after making a survey on legal recreational drugs and illegal drugs commonly used in your locality.

Drug Addiction

Drug addiction is a condition in which a person become dependent after using a drug again and again. Once a person gets addicted to a drug he/she cannot manage to work and engaged in daily routine without using it.
Therefore, it is advisable not to use any form of drug by any means unless it is prescribed for you by a physician.
After getting addicted to a drug it is very difficult to stop its usage suddenly. This may cause unpleasant effects on the body. These unpleasant effects are called**withdrawal symptoms**.
The withdrawal symptoms include:

- ✓ Unwell feeling
- ✓ Aches and pains
- ✓ Shaking
- ✓ Sweating
- ✓ Headaches
- ✓ Fevers
- ✓ Nausea and depression etc.

Types of Drugs

Based on their effect on our nervous system, drugs are classified into four types. These are:

- ✓ Stimulant
- ✓ Sedatives
- ✓ Analgesis (pain killers)
- ✓ Hallucinogens

i) Stimulants

Stimulants are drugs that increase the activity of the nervous system. They reduce feelings of fatigue and sense of alertness temporarily.

Stimulant drugs can be:

- ✓ **Mild stimulant** – are drugs that have stimulating effect on the body at smaller scale.

  Examples: Caffeine in tea, coffee, soft drinks
- ✓ **Heavy stimulant** – are drugs that stimulate the body at larger scale in unusual manner.

  Examples: Cocaine, khatonine

ii) Sedatives (depressants)

Sedatives are drugs that slow down the normal/proper working or functioning of the nervous system.
Examples: Alcoholic drinks, sleeping pills

> **Sleeping pills** are drugs that make people sleep. However, it is not recommended to use without prescription of a physician.

iii) Analesis (Pain killers)

Analgesis are drugs that dull our senses by affecting part of the nervous system that produces sense of pain.
Examples; Aspirin, paracetamol, etc.

iv) Hallucinogens

Hallucinogens are drugs that change the way we perceive things. They make the user to see and think about things that do not exist at all.
Examples: Cannabis (marijuana), LSD (lysergic acid)

Some examples of drugs and their impacts

i) Cigarette smoking

Cigarette consists of a powerful addictive chemical called **nicotine**. It also consists of more than 4,000 types of chemicals that have severe health effects on the user (smoker).
Some of the impacts of health caused due to smoking cigarette are:

- ✓ **Coronary heart disease** – is a condition where the walls of arteries gets narrower and reduce the supply of blood to heart and other parts of the body.

- ✓ **Arthrosclerosis** – damage on smooth lining of arteries.
- ✓ **Stoke** – the blockage of blood vessels in the brain or bleeding in the brain. Such condition may result in death.
- ✓ **Caners** – several types of cancers such as carce of limp, mouth, throat, pancreas, bladder, substances in cigarette.

Carcinogenic substances are cancer causing substances.

In addition to severe health problems, smoking cigarette has severe economic, hygienic and psychosocial consequences on the user and people around the smoker.

Smoking cigarette is not allowed in public areas, because the smoke exhaled (breathed out) from the user also affects other people around.

Even though cigarette has a powerful addictive nature, smoker should stop smoking by using different mechanisms in order to keep their health and live more.

Alcohol

Alcoholic drinks are used widely by people for recreational purposes. However, regular and intensive consumption of alcoholic drinks lead to addiction and health complications.

Alcohol is poisonous (toxic) and it can cause severe physiological, psychological and economic crisis on people when they get addicted to it.

Some of the health complications caused as a result of excessive consumption of alcohol are:

- ✓ Headaches, dehydration, nausea
- ✓ Lack of judgment
- ✓ Cirrhosis of the live – is the disease of the liver tissue which is likely to cause cirrhosis of the liver, lung cancer and even death.
- ✓ Disk of HIV/AIDS and unwanted pregnancy

Excessive consumption of alcohol leads people to risks of:

- ✓ domestic violence
- ✓ car accident
- ✓ Divorce, etc.

- ✓ **Arthrosclerosis** – damage on smooth lining of arteries.
- ✓ **Stoke** – the blockage of blood vessels in the brain or bleeding in the brain. Such condition may result in death.
- ✓ **Caners** – several types of cancers such as carce of limp, mouth, throat, pancreas, bladder, substances in cigarette.

Carcinogenic substances are cancer causing substances.

In addition to severe health problems, smoking cigarette has severe economic, hygienic and psychosocial consequences on the user and people around the smoker.

Smoking cigarette is not allowed in public areas, because the smoke exhaled (breathed out) from the user also affects other people around.

Even though cigarette has a powerful addictive nature, smoker should stop smoking by using different mechanisms in order to keep their health and live more.

Alcohol

Alcoholic drinks are used widely by people for recreational purposes. However, regular and intensive consumption of alcoholic drinks lead to addiction and health complications.

Alcohol is poisonous (toxic) and it can cause severe physiological, psychological and economic crisis on people when they get addicted to it.

Some of the health complications caused as a result of excessive consumption of alcohol are:

- ✓ Headaches, dehydration, nausea
- ✓ Lack of judgment
- ✓ Cirrhosis of the live – is the disease of the liver tissue which is likely to cause cirrhosis of the liver, lung cancer and even death.
- ✓ Disk of HIV/AIDS and unwanted pregnancy

Excessive consumption of alcohol leads people to risks of:

- ✓ domestic violence
- ✓ car accident
- ✓ Divorce, etc.

Domestic alcohol drinks that are widely used in Ethiopia are **tej, tella, katikala, etc**

Binge drinking – is consumption of excessive amounts of alcohol in a short period of time that leads of group violence.

Khat (ጫት)

Khat is a domestic shrub which leaf is chewed. It contains a stimulant drug called **khatonine**. Khat chewing seems to be a big problem in Ethiopia, since most of the productive and youth society of the country are getting addicted to it.

Chewing khat may cause loss of appetite and abdominal discomfort (unexplained).

Cannabis (Marijuana)

Cannabis is a dangerous hallucinant drug that contains chemicals called **caanabinoids**.

- ✓ **Delta – g – tetra hydro cannabinoid (THC)** is the most potent chemical that affect the brain cells responsible for memory, emotion and motivation.

Cannabis is illegal drug in Ethiopia.

Other drugs

- ✓ **LCD (Lysergic acid diethylamide)** is a very stronger hallucinogen that is made in the laboratory.
- ✓ **Cocaine** is a highly powerful stimulant during that causes depression in the end.
- ✓ **Herion**– is medially important painkiller that is made up of opium oppy. If used without prescription, it increases the risk of liver disease such as hepatitis, heart and brain damage.

Drug Abuse in Ethiopia

Drug abuse is the increasing public health problem in Ethiopia. The most commonly used drugs in Ethiopia are alcohol, khat and tobacco. However, there are illegal drugs that are used in Ethiopia. These are:

- ✓ Cannabis (marijuana) and solvents like gasoline are commonly used.

Illegal drugs that are rarely used in Ethiopia are:

- ✓ LSD (Lysergic acid)
- ✓ Estasy
- ✓ Cocaine
- ✓ Heroin

Both legal and illegal drugs have wide range of health problems on the user. However, illegal drugs cause **severe health problems** and **economic crisis** at individual and family level.

The commonly used recreational drugs in Ethiopia have the following features:

- ✓ they are addictive
- ✓ the affect the brain function and other behavior
- ✓ they damage health, resulting in lower productivity and absence from school or work.
- ✓ they adversely affect the individual, family, society and country.

Illustrative Example

39. When a person drinks coffee, it most likely makes him alert. This is because coffee contains:
    A. Stimulus
    B. Hallucinogens
    C. Sedatives
    D. Pain

    Explanation: A coffee consists a mild stimulant drug knownas caffeine. It makes as the user alert if used. **Answer: A**

40. Suppose a person takes unknown drug and becomes restless and less things that do not exist. The drug is most probably be:
    A. Pain killer
    B. Hypotonic
    C. Hallucinogen
    D. Stimulant

    Answer: C

41. Extremely powerful stimulant drug the ends in depression is:
    A. Cocaine
    B. Caffeine
    C. Lysergic acid
    D. Khatonine

Explanation:
- ✓ **Cocaine** is extremely powerful stimulant drug that ends in depression.
- ✓ **Caffeine** is a mild stimulant found in coffee, tea and soft drinks.
- ✓ **Lysergic acid** is a hallucinate drug in the laboratory.
- ✓ **Khatonine** is a stimulant in khat.

Answer: A

42. Which of the following NOT legal recreational drug in Ethiopia?
A. Khat
B. Coffee
C. Heroin
D. Alcohol

Explanation: Khat, coffee and alcoholic drinks are legal recreational drugs in Ethiopia. **Answer: C**

43. A strong hallucinant drug that is prepared in the laboratory is:
A. LSD
B. Cocaine
C. Cannabis
D. Marijuana **Answer: A**

44. The drugs are known to cause vivid walking dreams are:
A. Stimulant
B. Hallucinogen
C. Sedatives
D. Depressant

Explanation: Hallucinogen a drug that makes people to see and feel things that do not exist in real. It causes a vivid walking dreams. **Answer: B**

45. Which of the following statement is correct about drug abuse? It
A. has not risk as long as it is taken in small does.
B. has no any adverse effects
C. Increase the health of the brain.
D. is becoming more of health problem.

Explanation: Drug abuse
- ✓ Has risks weather it is taken in a small or large doses.
- ✓ Has adverse effects on health, economic and psychological state of the user.

Answer: D

46. Discuss the meanings of:
A. Illegal drug
B. Withdrawal symptom
C. Prescribed drug

Explanation:
- **Illegal drug** – is a drug that has adverse effect physiological, psychological and economic state of the user, family, society

and country. They are not legally allowed to be used, carried, transported or sold.

- **Withdrawal symptom** – unpleasant feeling caused due to sudden stopping of drug after addiction.
- **Prescribed drug** – is a kind of drug that is aided or recommended by a physician.

Post Test – 4

I. Choose the best option from the given alternatives.

1. Which of the following is illegal drug in Ethiopia?
   A. Caffeine C. Alcohol
   B. Khat D. Cocaine
2. Which of the following is a heavy stimulant?
   A. Cocaine C. Marijuana
   B. Caffeine D. LSD
3. Unpleasant feeling upon sudden stopping of drug is:
   A. Withdrawal symptoms C. Drug addiction
   B. Drug abuse D. Headache
4. A kind of drug that slows down the proper working of a nervous system is:
   A. Hallucinogen C. Sedatives
   B. Stimulant D. Analgesis
5. Sleeping pill is an example of:
   A. Stimulant C. Analgesis
   B. Sedatives D. Hallucinogen
6. A drug that cause a vivid walking dream is:
   A. Sedatives C. Stimulant
   B. Hallucinogen D. Analgesis
7. Which of the following drugs is hallucinogen?
   A. Cannabis C. Khatonine
   B. Asprin D. Sleeping pills
8. The powerful addictive chemical found in a cigarette smoke is:
   A. Khatonine C. Opium
   B. Nicotine D. Estacy
9. A drug that largely affect the liver is:
   A. Cigarette C. Alcohol
   B. Khatonine D. Cocaine
10. An excessive consumption of alcohol that leads to group violence is called:
    A. Group drinking C. Alcohol addictor
    B. Alcoholic D. Binge drinking

Part II: Match the following types of drug with their descriptions.

Column 'A'	Column 'B'
1. Stimulant	A. Dulls the sense
2. Sedatives	B. Slows down the working of nervous system
3. Analgesis	C. Increases the functioning of nervous system
4. Hallucinogen	D. Causes 'vivid walking dream'

The Sense Organ

Sense organs are organs the body that helps us to react to changes (stimuli) in and outside the body.
Sense organs consists of **sensory cells** which can detect a particular stimulus. This allows an organism to be aware of changes in the internal and external environment.

The Sensory receptors (sensory cells) are captable of changing energy of the stimulus in to electrical energy is nerve impulse.

Sense organs	Sensory receptors	Stimulus (energy)	Response
Eye	Cones and rods in retina	Light	Vision (seeing)
Ear	Cohlea	Vibration of sound (kinetic)	Hearing movement
Tongue	Taste buds	Chemicals in the form of solution	Taste
Nose	olifactory tissue	Chemicals in gaseous form	Smell
Skin	Various receptors • Thermo receptors • Thermo receptors	- Heat - Cold - Touch - Pressure - Pain	Temperature (Hotness or coldness)
Muscle	Stretch receptors		Movement

The eye

Eye is a spherical shaped sense organ that contains large number of sensory receptors. Human eye helps to see clear, three dimensional and colored image.

The overall structure of the eye is enclosed in a structure called **eye ball**. The eye ball is set or embedded in a hallow part of a skull called **socket**.

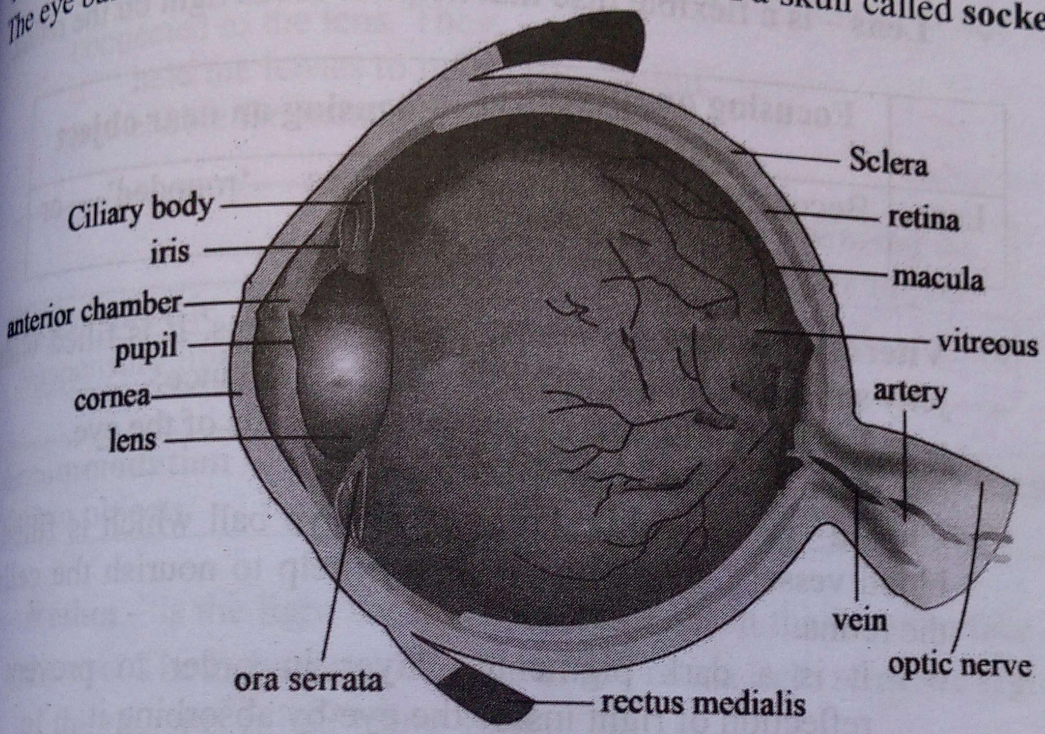


Figure 9.7: Structure of the eye

The eye is a very specialized and most sensitive sense organ. Thus, it is protected by various structures. These are:

- ✓ **Skull** or **cranium** – protects the eye from mechanical damages.
- ✓ **Eylid, eyelash,** and **eye burrow** – prevents the entrance of dusts to the eye.
- ✓ **Conjunctiva** – is a thin transparent membrance on the surface of the eye which prevents entrance of substances such as dust.
- ✓ **Tear gland** – secretes a fluid called **tear**. Tear helps to wash the surface of the eye.

Parts of the eye

Human eye has various parts that perform a specific task to enable us see. These parts are:

- ✓ **Scelera**– the outer layer of the eye ball. It provides protection of the eye.

- ✓ **Corena**– is the continuation of the scelerotic layer. It is a curved surface found on the front part of the eye. This helps the eye to bend the light rays inwards (into) the eye ball.
- ✓ **Aqueous humor** – is the fluid filled portion found in front of the lens. It supplies nutrients to the lens and corena. It also helps maintain the shape of the eyeball.
- ✓ **Lens** – is a flexible disc that helps to focus light on the retina.

	Focusing on distant object	Focusing on near object
Lens	Becomes 'flat' or thin	Becomes 'rounded' or thick

- ✓ **Viterosu humor** – is a part behind the lens. It is filled with a jelly substance having the following importance:
  - Provides nutrients (nourished) the cells of the eye.
  - Maintains the shape of the eye ball.
- ✓ **Chroid**– is the middle layer of the eye ball which is filled by blood vessels. These blood vessels help to nourish the cells of the retina.
  - It is a dark pigmented layer in order to prevent the reflection of right inside the eye by absorbing it.
- ✓ **Iris** – is the colored (pigmented) portion of the eye. It plays a role in controlling the amount of light entering into the eye ball.
- ✓ **Pupil** – is a hole in the center of the iris. It changes its size in response to changes in lighting.
  - They are two sets of muscles on the iris. These are **circular muscles** and **radial muscles**. They have antigonostic activities.
  - When radial muscles contract, circular muscles relax and vice versa.

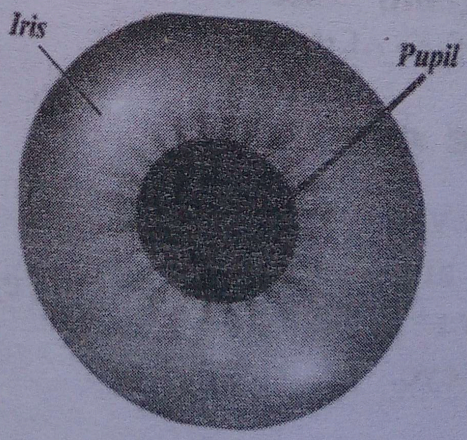


Figure 9.8: Iris and Pupil

	In bright light	In dim light
Pupil	Constricted	Dialates
Circular muscles	Contact	Relax
Radial muscle	Relax	Contract

✓ **Sensory ligaments** and **ciliary muscles** are structures that are connected to the lens. They
- held the lenses to held in a position.
- enable the lens to change its shape.

	Focusing on distant object	Focusing on near object
ciliary muscle	Relax	Contact
Suspensory ligament	Pulled tight	Slack

Accommodation – is the ability of the eye to see near and distant (far) to objects.

✓ **Retina** – is the light sensitive tissue lining in the inner surface of the eye. The light sensitivecells of the retina light sensitive region of the retina is called **blind spot**.
- **Yellow spot** – consist of two types of cells. These are **cones** and **rods**.
  - **Cones** – are cells in the retina that are sensitive to bright light.
  - **Rods**– are cells of the retina that are sensitive to dim light.

- The pigments in the rod are made up of **vitamin – A**. Thus, the lack of vitamin A results in poor vision during the night time where the light is dim or dark.
- The cone cells respond to red, green and blue light. Thus, color vision is the combination of these colors.

Focusing Light

For a clear image to be formed, an image must be focused on the **retina**. This is maintained by the association of the functioning of different parts of the eye such as:

- ✓ Bending (refracting) effects of the **cornea** and the **lens**.
- ✓ Regulating capacity of the pupil.
- ✓ Transparency of aqueous humor and vitreous humor.

When the light gets focused on the retina, an **inverted** and **real image** is formed. This is corrected by the vision center in the brain called **optic lobe**. An impulse is transported between the retina and the vision center in the brain by means of neurons called **optic nerves**.

The pathway of light and impulse to maintain our vision is:

Light passes through cornea → through aqueous fluid → through lens → through vitreous humor → image forms on retina → optic nerve transmit signals to the brain → vision in maintained.

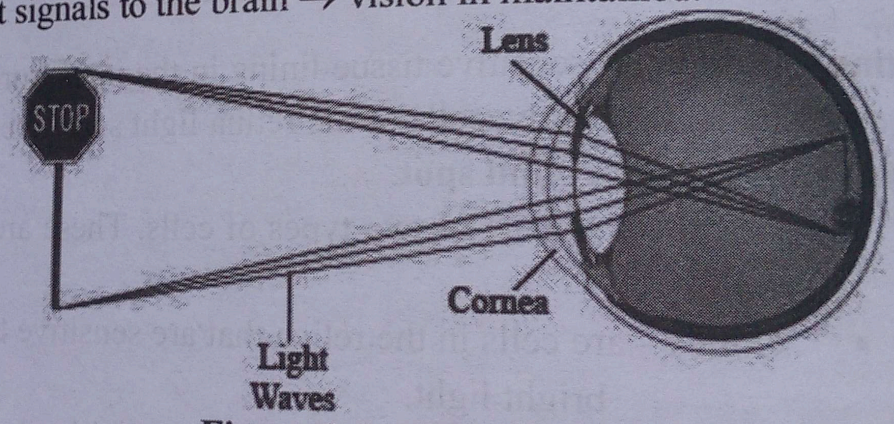


Figure 9.9: Focusing Light

Common defects of the eye

The normal functioning of the eye can be malfunctioned by various factors. These are called defects of the eye. Some defects of the eye may cause blinds (inability to see).

The most common defects of the eye are:

- ✓ Short sight
- ✓ Long sight
- ✓ Astigmatism

- ✓ Trachoma
- ✓ Cataract

i) Short sight (Myopia)

Short sightedness is a condition where a person or an individual can clearly focus on near objects but have difficulty of seeing distant (far) objects clearly.

In other words a person sees a blurry image when focusing on far (distant) objects.

Short sightedness is caused due to the following abo normal situations:

- ✓ When the eye ball is longer than its normal size.
- ✓ When the lenses of the eye is too strong or rigid.

Due to the above mentioned abnormalities a person with short sight problem faces a problem in which light is focused in front of the retina.

The defect of short sight is corrected by:

- ✓ Wearing eye glasses that are made up of **concave (diverging) lenses**.

Concave lens are lenses that bend the light says out wards.

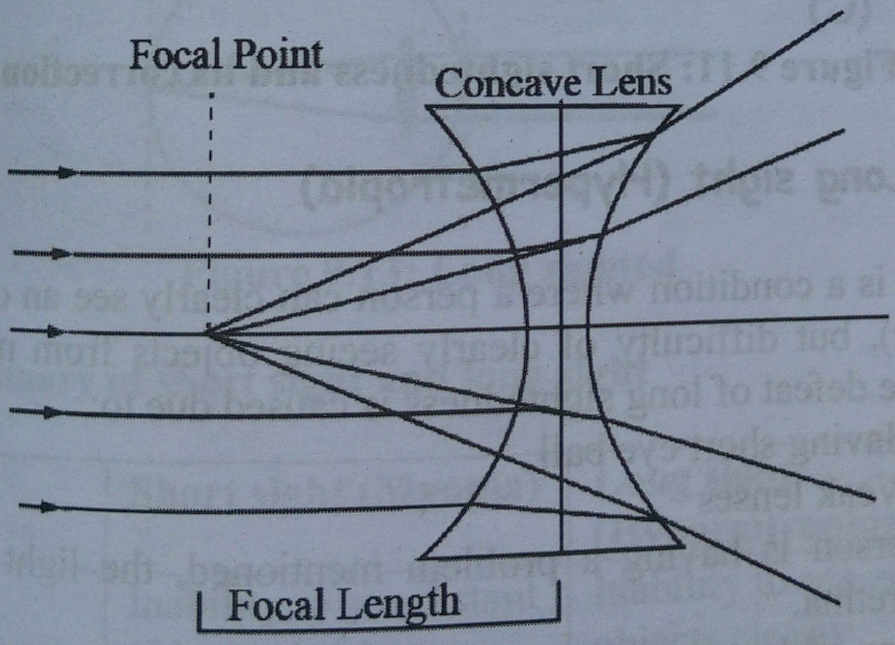


Figure 9.10: Concave lens

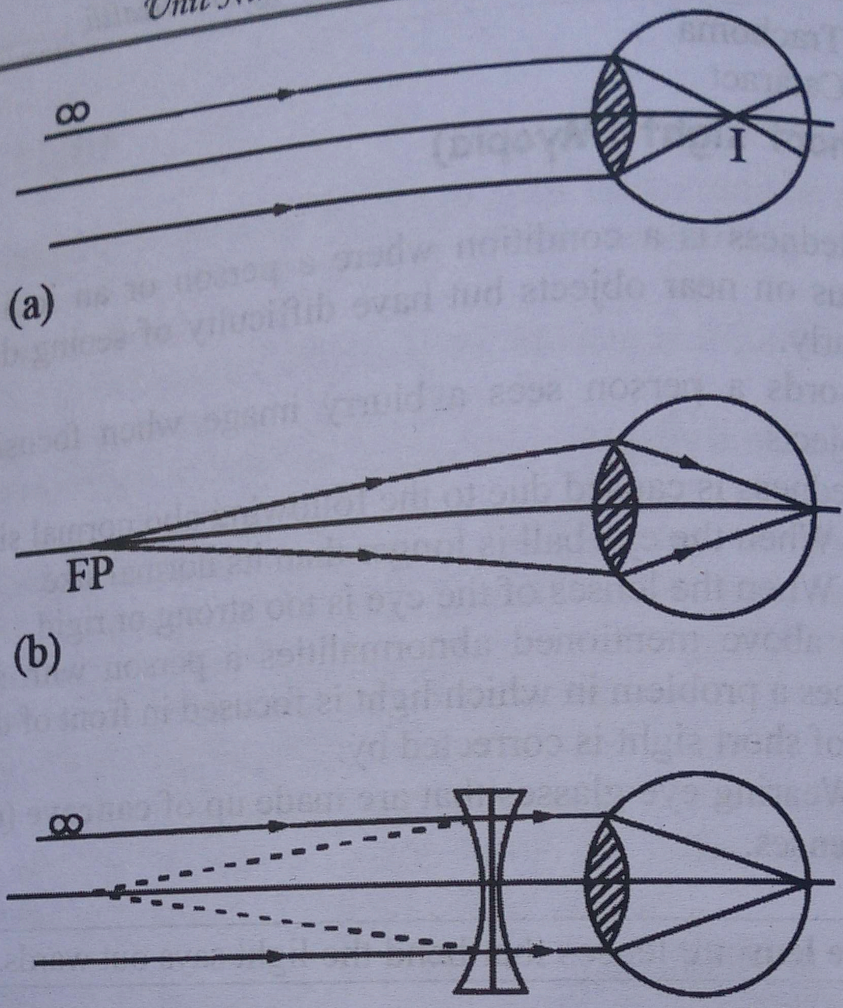


Figure 9.11: Short sightedness and its correction

ii) Long sight (Hypermetropia)

Long sight is a condition where a person can clearly see an object from distant (far), but difficulty of clearly seeing objects from near (close) objects. The defeat of long sightedness is caused due to:

- ✓ Having short eye ball
- ✓ Weak lenses

While a person is having a problem mentioned, the light is focused behind the retina.

The problem of long-sightedness is corrected by wearing eye glasses made from **convex (converging lenses)**.

Convex lenses have an ability to bend the light rays inwards.

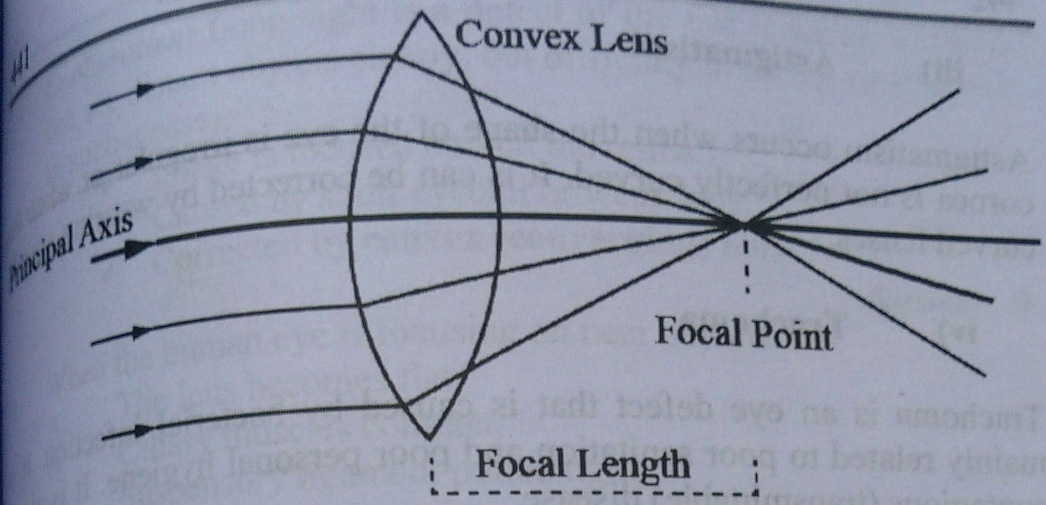


Figure 9.12: Convex lens

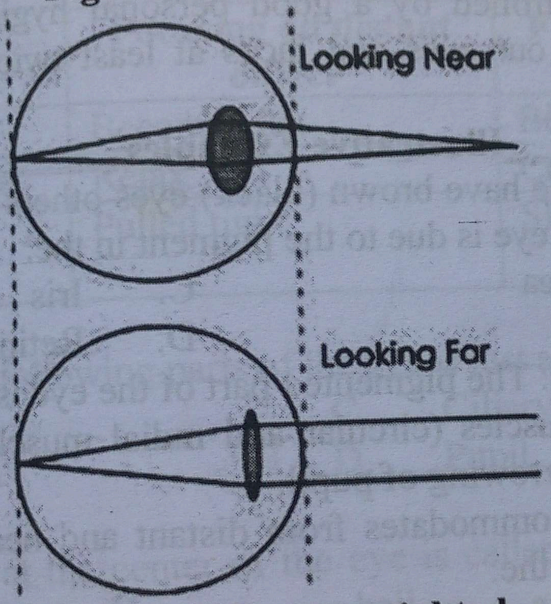


Figure 9.13: Long sighted

Table: Summary of short sight and long sight

	Short sight (Myopia)	Long sight (Hypermtrophia)
Defect	Inability to see distant objects clearly	Inability to see near objects clearly
Cause	• Long eye ball • Rigid or strong lens	• Short eye ball • Weak lens
Correction	Wearing concave (diverging) lens	Wearing convex (converging) lens

iii) Astigmatism

Astigmatism occurs when the shape of the eye is irregular or when the cornea is not perfectly curved. It is can be corrected by wearing equally curved lenses.

iv) Trachoma

Trachoma is an eye defect that is caused by bacterial infection. It is mainly related to poor sanitation and poor personal hygiene. It is also a contagious (transmittable) disease.
If not treated timely or properly, trachoma may cause blindness.
Trachoma is controlled by a good personal hygiene. Some of thes include, washing our eyes and faces at least twice a day and prope disposal of wastes.

Illustrative Examples

47. Some people have brown (black) eyes others have blue eyes. Th color of the eye is due to the pigment in the:
    A. Cornea
    B. Pupil
    C. Iris
    D. Retina

    Explanation: The pigmented part of the eye is iris. It also consi of set of muscles (circular and radial muscles) that control t widening narrowing of pupil. **Answer:**

48. The eye accommodates from distant and near objects primar by changing the:
    A. Size of pupil
    B. Shape of lens
    C. Length of the lens
    D. Distance of the retina from the lens

    Explanation: Our eye can focus on objects from near and dista due to the elastic and flexible nature of the lens. This abilit called**accommodation**. **Answer**

49. Which of the following is incorrect about a long sighted perso
    A. Can see distant objects clearly.
    B. Corrected by wearing concave lens.
    C. Image is focused behind the retina.
    D. Caused due to short eye ball.

Explanation: Long sight is a defect of the eye is which a person can see distant objects clearly, but difficulty in seeing near object. In such a case:

- ✓ Image is focused behind the retina.
- ✓ Caused by short eyeball or weak lens.
- ✓ Corrected by **convex (converging) lens.**

Answer: B

50. When the human eye is focusing on near objects:
A. The lens becomes flat.
B. Ciliary muscles contract.
C. Suspensory ligament pulled tight.
D. Pupil dilates

Explanation:

	Focusing on distant object	**Focusing on near object**
Lens	Becomes flat	Becomes rounded
Ciliary muscle	Relax	Contact
Suspensory ligament	Pulled tight	Slack

Answer: B

51. Which of the following parts of the eye is just an opening?
A. Iris
B. Cornea
C. Ciliary body
D. Pupil

Explanation:
The opening at the center of the eye is called**pupil**. It regulates the amount of light that enters eye ball.

- ✓ In bright light pupil constricts.
- ✓ In dim light pupil dilates.

Answer: D

52. If you observe a person holding the paper at distance from his eye in order to read it, which of the following eye defects is he/she most probably suffering by?
A. Astigmatism
B. Color blindness
C. Short sight
D. Long sight

Explanation:
This person is holding paper at distance, because she/he has a difficulty of seeing objects from near. Thus, the defect of this person is long sight.

Answer: D

53. People having an eye defect of short sight cannot see distant objects clearly because:
A. The eye forms a very short image of the object.
B. The light is focused on the blindspot.
C. The light is focused on the fouea.
D. The light is focused in front of the retina.
Explanation: Short sighted person cannot clearly see distant objects because the light is focused in front of the retina.
Answer: D

54. When Bekele looks at a distant object, his eyes form a blurred image. As the object gets closer and closer to him the image becomes clearer and clearer. Bekele's eye defect is called:
A. Long sight
B. Short sight
C. Astigmatism
D. Night blindness
Explanation: If a person cannot clearly see an object from distant but clearly see a near object, the defect of this person is known as **short sight** or **myopia**.
Answer: B

55. The function of ciliory muscles of the eye is:
A. Controlling the shape of the lens.
B. Controlling the size of the pupil.
C. Rotating the eyeball in the socket.
D. Controlling the shape of the eye.
Explanation: Ciliary muscles are responsible for controlling the movement and shape of the lens.

	While focusing on distant object	**While focusing on near object**
Ciliary muscle	Relax	Contact
Lens	Flat	Rounded

Answer: A

56. If a person is short sighed, what kind of eye glasses would you advice him/her to wear? Eye glasses with:
A. Concave lens
B. Convex lens
C. thin lens
D. Thick lens
Explanation: Short sight defect of the eye is corrected by wearing **concave lens**.
Answer: A

57. Distinguish between blind spot and yellow spot.
Explanation:
- ✓ Blind spot is the part of the retina where light sensitive cells are not found.
- ✓ Yellow spot is the part of the retina where the light sensitive cells; cones and rods are found.

53. People having an eye defect of short sight cannot see distant objects clearly because:
A. The eye forms a very short image of the object.
B. The light is focused on the blindspot.
C. The light is focused on the fouea.
D. The light is focused in front of the retina.

Explanation: Short sighted person cannot clearly see distant objects because the light is focused in front of the retina.

Answer: D

54. When Bekele looks at a distant object, his eyes form a blurred image. As the object gets closer and closer to him the image becomes clearer and clearer. Bekele's eye defect is called:
A. Long sight
B. Short sight
C. Astigmatism
D. Night blindness

Explanation: If a person cannot clearly see an object from distant but clearly see a near object, the defect of this person is known as **short sight** or **myopia**.

Answer: B

55. The function of ciliory muscles of the eye is:
A. Controlling the shape of the lens.
B. Controlling the size of the pupil.
C. Rotating the eyeball in the socket.
D. Controlling the shape of the eye.

Explanation: Ciliary muscles are responsible for controlling the movement and shape of the lens.

	While focusing on distant object	**While focusing on near object**
Ciliary muscle	Relax	Contact
Lens	Flat	Rounded

Answer: A

56. If a person is short sighed, what kind of eye glasses would you advice him/her to wear? Eye glasses with:
A. Concave lens
B. Convex lens
C. thin lens
D. Thick lens

Explanation: Short sight defect of the eye is corrected by wearing **concave lens**.

Answer: A

57. Distinguish between blind spot and yellow spot.

Explanation:

- ✓ Blind spot is the part of the retina where light sensitive cells are not found.
- ✓ Yellow spot is the part of the retina where the light sensitive cells; cones and rods are found.

58. Write the functions of the following parts of the eye.
✓ Pupil
✓ Lens
✓ Retina
✓ Optic nerve

Explanation:
✓ **Pupil** – regulates the amount of light entering the eye ball.
✓ **Lens** – helps to see near and distant objects.
✓ **Retina** – is the part of the eye where image of formed.
✓ **Optic nerve** – carries impulse from the retina to the vision center in the brain.

Post Test 5

Part I: Choose the best option from the given alternatives.

1. Which of the following sense organs is incorrectly matched with the kind of stimulus detected by their receptors?
   A. Eye – light
   B. Ear – mechanical
   C. Tongue – Chemical
   D. Nose – Mechanical
2. All of the following are the protective structures of the eye except one:
   A. Cranium
   B. Eye lid
   C. Conjunctive
   D. Vertebrae
3. The most convex part of the eye is:
   A. Sclera
   B. Cornea
   C. Conjunctiva
   D. Lens
4. The amount of light that enters the eye ball is regulated by
   A. Cornea
   B. Pupil
   C. Lens
   D. Retina
5. While focusing on distant object,
   A. The lens becomes flat and thin.
   B. The lens becomes thick and rounded.
   C. The lens becomes thin and rounded.
6. In dim light:
   A. Pupil gets constricted.
   B. Circular muscles contract
   C. Radial muscles contract
   D. Radial muscles relax
7. The ability of the eye to see near and distant objects is called:
   A. Focusing
   B. Accommodation
   C. Seeing
   D. Observation

8. The cell of retina that are sensitive to dim light are:
   A. Rods
   B. Blind spots
   C. Cones
   D. Optic lobe

9. Which one of the following is correct concerning short sighted problem?
   A. The eye ball is shorter than its original size.
   B. It is a difficulty of seeing near objects.
   C. It is corrected by concave lens.
   D. Image tends to be focused behind the retina.

10. Which one of the following eye defect is caused by germs?
   A. Astigmatism
   B. Trachoma
   C. Short sight
   D. Long sight

Part II: Fill in the black spaces using appropriate answer.

1. The curved and most convex part of the eye is _____.
2. The cells of the retina that defect bright light are _____ and dim light are _____.
3. The colored part of the eye is called_____.
4. _____are nerves that carry an impulse from the retina of the eye of the brain.
5. _____is the ability of the eye the see near and distant objects.

Part III. Matching

Column A	Column B
1. Short sight	a. Bends light rays outwards
2. Astigmatism	b. The colored part of the eye
3. Trachoma	c. The most convex layer of the eye
4. Long sight	d. The cells that detect bright light
5. Converging lens	e. Ability of the eye to see near and far objects.
6. Diverging lens	f. Bends light rays inwards
7. Rod	g. The cells that detect dim light
8. Optic nerve	h. The hole at the center of the iris

9. Iris	i. Eye defect caused by bacterial infection
10. Cone	j. Carries an impulse from retina to the vision center in the brain
11. Cornea	k. Eye defect that is caused when the shape of eye ball is irregularly shaped.
12. Pupil	l. Difficulty of seeing near objects
13. Accommodation	m. Corrected by concave lens

2) The ear

The ear is an organ for hearing and balance. It has the following major parts;

- ✓ Outer ear
- ✓ Middle ear
- ✓ Inner ear

i) The Outer ear

The outer ear has the following main parts:

- ✓ **Pinna** – is a cartilegenous flap that serves to:
  - o Collect sound vibrations
  - o Know the direction of the sound.
  - o Regulate temperature of the body (morphological means)
- ✓ **Ear Canal** – is a 2cm long tube that carries the sound in wards from pinna.
- ✓ **Ear drum (Tympanium)** – is a memraneosu structure that forms vibrations of the incoming sound.

ii) The Middle ear

The middle ear consists of tiny bones called **ossicles**. The ossciles are three in number and they are:

- ✓ Hammer (Malleus)
- ✓ Anvil (incus)
- ✓ Stirrup (stapes)

Ossicles are the smallest bones in the entire body.

The ossicles are used by **amplify** the vibration of sound.
Eustachian tube is a tube that connects the middle ear with the throat. It helps to balance (equalize) the pressure in the ear.

iii) Inner ear

Inner ear is fluid filled portions that consist of sac – like structures called **sacuullus** and **utricles, three semi circular canals** and **cochlea**.

- ✓ **Semi circular canals** are structures that consist of a swollen portion at the base called **ampullae**. Ampullae can detect the movement of fluids with in the canal semicircular canals are responsible for detecting the direction of motion.
- ✓ **Cochlea** – consists of hair like sensory cells that can detect the vibration of sound. These sensory cells are responsible for our hearing. The pathway of vibration in the ear is:

Pinna → ear canal → ear drum → malleus → incus → stapes → round window → cochlea → auditory nerve → brain

An impulse is carried from inner ear to be brain by means of **auditory nerves**.

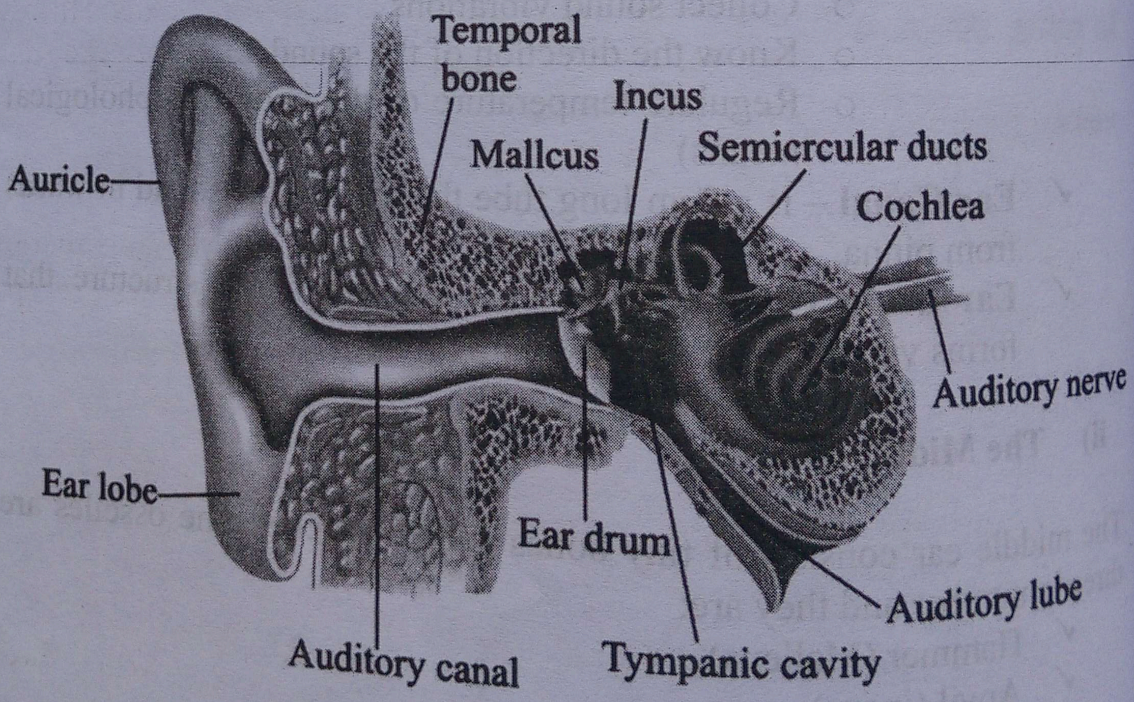


Figure 9.14: The human ear

Common Disorders of the Ear

The proper functioning of the human ear can be affected by various causes. Some of these are:

- ✓ Mechanical damage – including very loud voices,
- ✓ Bacterial infection
- ✓ Age

The disorders of the ear may result in **deafness**. Deafness is inability to hear. It can be temporary or permanent.

- Temporary deafness can be caused by very loud voices. It may end in permanent deafness.
- Permanent deafness results due to the following factors. These are:
  - Infection of the middle ear
  - Age
  - Damage to auditory nerve
  - Disease that may be caused by infection.

3) Sense of taste and Smell

The sense of taste and smell are detected by different organs. However, both have one major thing in common. They are receptors that detect chemical stimuli. Thus, they are called**chemoreceptors**.

i) Taste

An organ for the sense of taste is **tongue**. On the tongue there are receptors that detect chemicals in the form of solution. These receptors are called**taste buds**.

The taste buds detect five types of tastes. These are:

- ✓ Sweet
- ✓ Bitter
- ✓ Sour
- ✓ Salty
- ✓ Umami

The receptors of the tongue (taste buds) are evenly distributed through the tongue. Thus, there are taste buds for all types of tastes across every region of the tongue regardless of the difference on their concentration in different regions.

Umami is a newly discovered sense of taste which is characterized by a very **savory flavor**. Such a flavor is found in food stuffs such as meat, cheese, broth and mushroom.

ii) The Sense of Smell

An organ for sense of smell is **nose**. The receptors for the sense of smell are **olfactory** epithelial cells. They detect chemicals in the gaseous state (in the form of vapor).

Our sense of smell is reduced when we suffer from cold of flu. This is because the smell receptors are covered with mucus.

The food we taste is also affected by what we smell.

Smell receptors are more sensitive than taste receptors.

Illustrative Example

59. Suppose 'person A' has functional smell and taste receptors, while, 'person B' has nonfunctional smell receptors and functional taste receptors. Person 'A' and 'B' are given the same type of food to taste. Who will taste the food better?
    A. Both will equally taste the food
    B. Person B
    C. Person A
    D. Both could not taste the food

    Explanation: The flavor of the food we eat is affected by both the senses of taste and smell. Thus, a person having a functional sense of taste and smell tastes the food better. **Answer: C**

60. Which of the following bones of the middle ear fits between other two bones
    A. Incus B. Striup C. Malleus D. Stapes

    Explanation: The sequence of ossicle bones is:
    malleus (hammer) → incus (anvil) → stages (stirrup)
    Thus, incus (anvil) fits in between malleus (hammer) and stages (stirrup). **Answer: A**

61. How is the pressure in the middle ear equalized to that in the throat when you are flying in an airplane?
    A. By shaking your heat continuously
    B. By opening the Eustachian tube
    C. By plugging cotton into your ear canal

D. By closing the pinna into your ear canal

Explanation: Eustachian tube comment the middle ear with the throat. When it opens, it balances or equalizes the pressure in the ear. **Answer: B**

62. Which of the following structure is different from the others?

A. Stapes B. Sclera C. Malleus D. Incus

Explanation: Stapes, malleus and incus are parts of the ear, while, scelera is part of the eye. **Answer: B**

63. Which of the following ear bones directly transmits sound to the cochlea?

A. Stapes B. Malleus C. Incus D.Ampullae

Explanation: Cochlea is the part of the inner ear and receives vibration from stapes (stirrup). **Answer: A**

64. What do we call the swelling on each semicircular canal of the human ear?

A. Capula C. Eardrum

B. Ampullae D. Eustachian tube

Explanation: Ampullae is the swollen portion at the base of semicircular canals. It detects the movement of fluids.

Answer: B

65. Which of the following structures of human ear collects sound waves?

A. Ear canal C. Eustachian tube

B. Cochlea D. Pinnae

Explanation:

Pinnae is a cartilaginous flap found at the most outer part of the ear. It is used to collect the sound waves.

Answer: D

66. Receptors that detect body position are found in the:

A. Eye B. Ear C. Skin D. Nose

Explanation: Ear is a sense organ for hearing and balance.

Answer: B

67. Which of the following structures is found in the fluid filled portion of the ear?

A. Ossicles C. Tympanic membrane

B. Semicircular canal D. Pinna

Explanation: The fluid filled portion of the ear is the inner . It consists of structures such as semicircular canal, cochlea utriallus and sacculus. Answ

68. Which of the following is the correct order in which vibrations caused by sound waves pass from the outer the inner ear?
A. Malleus → Incus → Stapes
B. Malleus → Stapes → Incus
C. Incus → Malleus → Stapes
D. Incus → Stapes → Malleus

Explanation: The correct order of ossicles from outer to inner ear is: Malleus (Hammer) → Incus (Anvil) → Stapes (stirrup)

Answer: A

69. What are the two bones of the middle ear that fit to the eardrum and the oval window respectively?
A. Stapes and anvil
B. Malleus and stapes
C. Mallevs and anvil
D. Stapes and incus

Explanation: Malleus (hammer) fits into the ear drum and stapes (stirrup) fits into the oval window.

Answer: B

70. This item is based on the following table.

Stimulus	Receptors
Light	Retina
X	Ampullae
Chemical	Tongue
Touch	Skin

Stimulus 'X' is:
A. Heat
B. Chemical
C. Movement
D. Sound

Explanation: The stimulus for movement is detected by a swollen portion at the base of semicircular canals called **ampullae**.

Answer: C

71. Which of the following senses are detected by the same organ?
A. Taste and smell
B. Hearing and balance
C. Sight and hearing
D. sight and smell

Explanation: Ear is an organ that is used for the senses of **hearing** and **balance**.

Answer: B

72. Which of the following ear structures collects sound waves directs them to the ear drum?
A. Pinna B. Middle ear C. Malleus D. Cochlea

Explanation:

Pinna is a cartilaginous flap that collects the sound waves to the ear drum.

Answer: A

73. Identify the following parts of the ear which are parts of the outer ear, middle ear and inner ear.

- ✓ Pinnae
- ✓ Cohchlea
- ✓ Ossicles
- ✓ ear drum
- ✓ semicircular canal
- ✓ pymphanic membrane

Explanation:

Outer ear	Middle ear	Inner ear
Pinnae, ear drum	Ossicles, tympanic membrane	Cochlea, semicircular canal

Post Test 6

I. Choose the best answer from the given alternatives.

1. The part of the ear where vibration of the incoming sound is formed is:
   A. Pinna
   B. Ear canal
   C. Ear drum
   D. Cochlea
2. Which of the following structure of the ear is involved in amplification of vibration of sound?
   A. Mallevs
   B. Incus
   C. Stapes
   D. All are possible answers
3. The correct order of ossicles in the middle ear is:
   A. Stapes → malleus → Incus
   B. Malleus → Stapes → Incus
   C. Malleus → Incus → Stapes
   D. None of the above
4. The pressure in the middle ear is equalized by:
   A. Ossicles
   B. Eustachian tube
   C. Ear canal
   D. Pinna
5. The part of the ear that contain receptors for hearing is:
   A. Ossicles
   B. Semicircular canal
   C. Cochlea
   D. Oval window
6. An impulse is carried from the inner ear to the brain by:
   A. Auditory nerve
   B. Olfactory nerve
   C. Optic nerve
   D. None of the above

7. Which of the following is not the function of the ear?
   A. Balance
   B. Posture
   C. Memory
   D. Hearing
8. Which of the following is not the main cause of permanent?
   A. Infection
   B. Damage to auditory nerve
   C. Age
   D. Very laud voice
9. Which of the following is attached to tympanum?
   A. Malleus
   B. Anvil
   C. Incus
   D. Stirrup
10. The newly discovered sense fo taste that is characterized by savory taste is:
   A. Sweet
   B. Umami
   C. Bitter
   D. Salty
11. Which of the following senses are detected by chemoreceptors?
   A. Smell and taste
   B. Vision and hearing
   C. Smell and balance
   D. Taste and touch
12. Receptors for the sense of taste are:
   A. Taste buds
   B. Olfactory cells
   C. Utricullus
   D. Sacculus

Matching

Column A	Column B
1. Pinna	a. Concerned with detection of the sense of smell.
2. Cochlea	b. Collects the sound waves into the ear.
3. Taste buds	c. Detects chemicals in the form of solution
4. Eustachian tube	d. Equalizers pressure in the middle ear
5. Olfactory cells	e. Contains cells for hearing in the inner ear

The Skin as a sense organ

Skin is the largest sense organ in the entire body. It has also number of important functions in the body. The skin is also called a **general sense organ** because it contains receptors for different senses. These are:

- ✓ Touch
- ✓ Pressure
- ✓ Temperature
- ✓ Pain

The senses stated above are detected by different types of sensory receptors. These are:

Senses	Receptors
Touch	Meissonerscorupuslces
Pressure	Pacinian corpuscles
Temperature	Thermo receptors
Pain	Nocireceptors

The skin has also other impotant functions. These are:

- ✓ Protection of the body from entrance of pathogens.
- ✓ Protection from damage by UV right to he sun.
- ✓ Excretory organ → removal of nitrogenous wastes, excess water an others in the form of sweat.
- ✓ Regulation of body temperature.
- ✓ Prevents excessive loss of water by evaporation.

The human skin has three major layers. These are:

- ✓ **Hypodermis** – is the lower fatty layer of the skin which helps to insulate the body against heat loss.
- ✓ **Dermis** – is middle layer of the skin, which consists of blood vessels, lymph vessels, hair follicles and sweat glands.
- ✓ **Epidermis** – is he outer most layer of skin, comprising dead cells which protects against water loss and entry of pathogens.

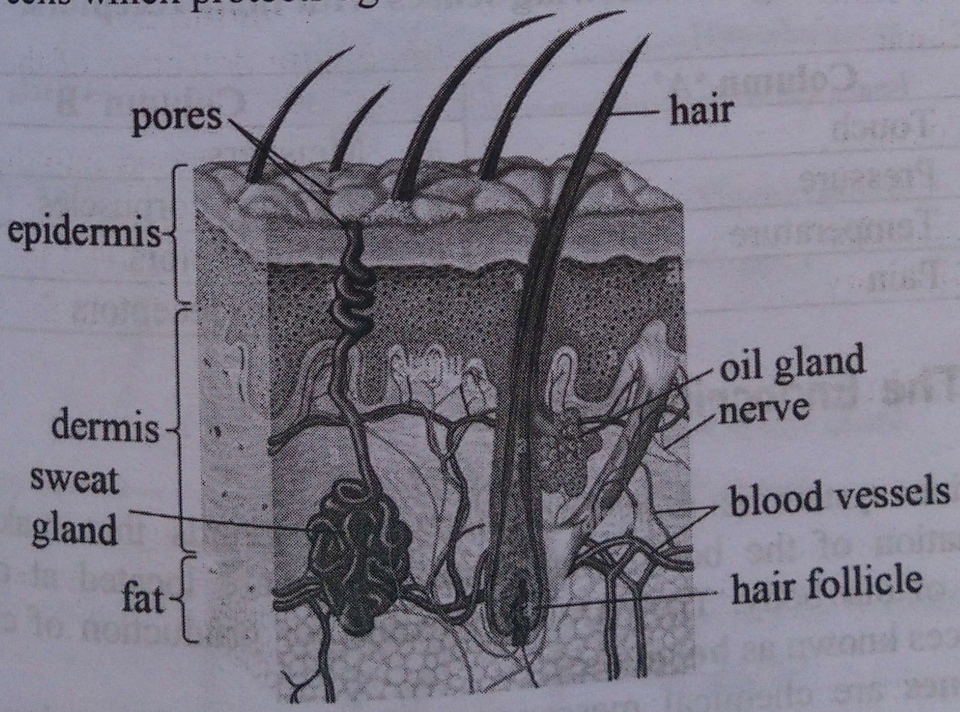


Figure 9.15: The cross sectional areas of the skin

Post Test 7

Part I: Choose the best answer from the given alternatives.

1. The protective layer of the skin is:
   A. Dermis
   B. Epidermis
   C. Hypodermis
   D. Blood vessels
2. Which of the following sense is NOT located on the skin?
   A. Touch
   B. Pressure
   C. Smell
   D. Pain
3. One of the following is the function of the skin, which one is it:
   A. Protection from pathogens
   B. Regulation of body temperature
   C. Damage by ultraviolet radiation
   D. All are possible answers
4. The fatty insulting layer of the skin is:
   A. Epidermis
   B. Hypodermis
   C. Dermis
   D. Hair Follicles
5. The layer of the skin which consists of blood vessels, lymph vessels, hair follicies and sweat glands is:
   A. Hypodermis
   B. Epidermis
   C. Dermis
   D. Hair follicles

Part II: Matching the following senses with their receptors.

Column 'A'	Column 'B'
1. Touch	a. Meisners
2. Pressure	b. Pacinian corpuscles
3. Temperature	c. Nocireceptors
4. Pain	d. Thermo receptors

9.2 The Endocrine Glands

Endocrine system is a system of ductless glands that make up a coordination of the body. Endocrine glands are located at different regions of our body. They are responsible for production of chemical substances known as **hormones**.

Hormones are chemical messengers of the body. They play a very significant role in facilitation of different processes in our body. Some of these are:

- ✓ Reproductive system
- ✓ Regulation fo blood sugar level.
- ✓ Regulation osmotic (water balance) of the body.
- ✓ Metabolism of oxygen, etc.

From these we can see that hormones have wide range of effects on our metabolic process.

Hormones act only on a specified organ called **target organ**. The effect of hormones on the body can stay for a prolonged period of time as compared to the nervous system. However, the way hormones act on the body is slower as compared to electrochemical system of the nervous control.

Endocrine glands are **ductless glands**. This means they are not directly connected to any system of duct or tube.But they simply, release that hormones into the blood stream by diffusion.

Glands are structures that secrete different useful substances such as hormones and enzymes. They can be classified as:

- ✓ **Exocrine glands** – are glands that secrete products other than hormone, such as enzymes, tear, saliva etc.
  Examples: pancreas, salivary glands, etc.
- ✓ **Endocrine glands** – are ductless glands that secrete hormones.

Examples: Thyroid gland, adrenal gland, pituitary gland, etc.

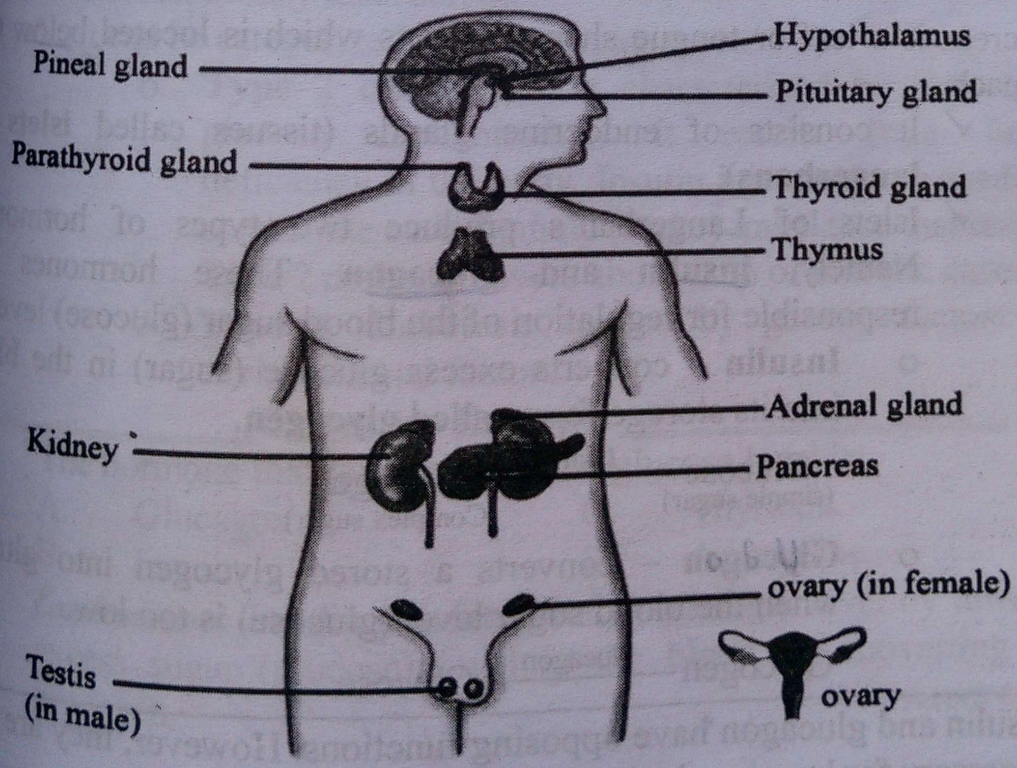


Figure 9.16: The position of the main endocrine gland sin human

The major endocrine glands of the human body are summarized in the following table:

Endocrine gland	Location	Hormones they secrete
Pancreas (Inslets of longer hands)	Below the stomach	Insulin and glucagon
Thyroid gland	Neck region (front)	Thyroxine
Adrenal gland	Top of each kidney	Adrenaline
Pituitary gland	Base of the brain	- ADH (anti-duretic hormone) - FSH (Follicle stimulating homrone) - LH (Leutenizing hormone) - Growth hormone
Testis	Scrotum of males	Testosterone
Ovary	Reproductive organ of female	Estrogen and progesterone
Parathyroid gland	Neck region (back)	Parathyroid hormone

1. Pancreas

Pancreas is a leaf or tongue shaped organs which is located below the stomach.

- ✓ It consists of endocrine glands (tissues called **islets of langerhans**).
- ✓ Islets of Langerhan's produce two types of hormones. Namely, **insulin** and **glucagon**. These hormones are responsible for regulation of the blood sugar (glucose) level.
  - **Insulin** – converts excess glucose (sugar) in the blood into its storage form called **glycogen**.

    Glucose (simple sugar) $\xrightarrow{\text{Inslin}}$ Glycogen (Complex sugar)
  - **Glycogen** – converts a stored glycogen into glucose when the blood sugar level (glucose) is too low.

    Glycogen $\xrightarrow{\text{glucagon}}$ Glucose

Insulin and glucagon have opposing functions. However, they are both necessary for keeping the body in a steady (normal) state.

The target organ for **insulin** and **glucagon** is the **liver**. Liver is a storage site of glycogen.

Do you remember?

The storage form of carbohydrate in animal is **glycogen;** while the storage form of carbohydrate in plants is **starch**.

Both glycogen and starch are complex sugars which are stable and insoluble forms of sugar in the body.

Pancreas acts as both as **endocrine gland** and **exocrine gland**.

- ✓ It is exocrine gland because it produces digestive enzymes that are known as **lipase, trypsin** and **amylase**.
- ✓ It is an endocrine gland because it secretes hormones **insulin** and **glucagon**.

Quick Task

- Distinguish between exocrine and endocrine glands.

Diabetes Mellitus

Diabetes Mellitus is one of the most common endocrine disorder. It is a serious health problem worldwide.

- ✓ It is a major cause of cardiovascular disease, blindness, nerve disease, kidney disorder, etc.
- ✓ It can be categorized into two major types.
  - **Type 2 diabetes** – is characterized by high blood glucose concentration. It results from insulin deficiency in the body. Insulin injections are needed to correct the carbohydrate (sugar) imbalance that results. Type 2 diabetes covers about 90% of diabetic cases.
  - **Type 1 diabetesis** caused by genetic factors and environmental factors.

Illustrative Example

74. The hormone that lowers the blood glucose level is:
    A. Glucagon
    B. Insulin
    C. Thyroxin
    D. Adrenaline

Explanation: Insulin regulates the blood sugar level by lowering excess sugar (glucose) level in the blood by converting into glycogen.

Answer: B

75. The two hormones that act antagonistically to one another in regulating the level of glucose in the blood are:
A. Gonadotrophin and progesterone
B. Androgen and glucagon
C. Insulin and glucagon
D. Glucagon and androgen

Explanation: Insulin and glucagon are two hormones that act antagonistically (opposing manner) to regulate the blood sugar level.

$$\text{Glucose} \underset{\text{Glucagon}}{\overset{\text{Insulin}}{\rightleftharpoons}} \text{Glycogen}$$

Answer: C

76. Diabetes mellitus is the result of underproduction of:
A. ADH
B. Aldosterone
C. Glucagon
D. Insulin

Explanation: Type 2 diabetes results when there is a deficiency of insulin. This results in the rise of glucose (sugar) concentration in the blood.

Answer: D

77. What is the role of liver in regulating the blood sugar level, when it rises above the normal?
A. Respiring the excess sugar in the body.
B. Excreting the excess sugar in the blood.
C. Converting the excess sugar into glycogen and store.
D. Converting the excess sugar in to protein and store.

Explanation: Liver is a target organ for insulin and glucagon where glycogen is stored.

Answer: C

78. Which of the following is not true about hormones?
A. They are transported by the blood.
B. They are produced by exocrine glands.
C. They act on target cells which lack receptors.
D. They have effect on all tissues and organs.

Explanation: Hormones are:
- ✓ Transported by the blood
- ✓ Produced by exocrine glands
- ✓ Act on target cells or organs.

Answer: D

79. How hormones do find their target cells?
A. The glands and the cells are located near to each other.
B. The cells contain receptor cells specific to the hormone.
C. The hormones are taken to the cells by carrier molecules.
D. The hormones and the cells are made up of identical materials.

Answer: B

80. What is the cause of diabetes mellitus?
A. More production of adrenaline by adrenal gland.
B. More production of progesterone by ovary.
C. Less or no production of insulin by pancreas.
D. Less or no production of bile by the liver.
Explanation: One of the cause of diabetus mellitus is due to less or no production of insulin by pancreas. **Answer: C**

81. Which of the following statement is correct about exocrine glands? They
A. are ductless
B. have duct
C. Secrete hormone directly into blood
D. Secrete enzymes directly into the blood.
Explanation: Exocrine glands have ducts and they secrete enzymes through their duct. **Answer: B**

82. Which of the following structures produces hormones?
A. Gland B. Neuron C. Organ D. Vessels
Explanation: Hormones are produced by ductless glands known as **endocrine glands**. **Answer: A**

83. Which of the following statement is true about exocrine gland? It
A. Has special tube to carry its secretion.
B. Is ductless and controls reproduction.
C. Secretes hormone directly into the blood.
D. Includes pituitary and thyroid glands.
Explanation: Exocrine glands:
✓ Have special tube to carry their secretions
✓ They have ducts **Answer: A**

84. What is the difference between endocrine and exocrine glands? Endocrine glands?
A. Are more in number in our body than exocrine glands.
B. Have ducts whereas exocrine glands do not have ducts.
C. Do not have ducts; whereas, exocrine glands have ducts.
D. Release their secretion through ducts to size of destination.
Explanation:
✓ Exocrine glands have ducts; while, endocrine glands are ductless.
✓ Exocrine glands release their secretions by means of ducts which they are connected with; while, endocrine glands directly release their secretions in to the blood stream. **Answer: C**

85. Which one of the following method is used to treat diabetes?
A. Eating red meat
B. Taking more glucose
C. Taking antibiotics
D. Insulin injection

Explanation: Diabetes can be treated by:
✓ Getting an insulin injection **Answer: D**

86. Diabetes mellitus can be treated by
A. Injecting insulin before meal
B. Avoiding contact with infected person
C. Avoiding sex with infected person
D. Keeping personal hygiene

Explanation: Diabetes mellitus can be treated by:
✓ Getting insulin injection before meal. **Answer: A**

87. Match the following endocrine glands with the hormone which they secrete.

Column A		Column B	
1.	Thyroid gland	a.	Estrogen
2.	Pancreas	b.	Insulin
3.	Adrenal gland	c.	Thyroxin
4.	Pituitary gland	d.	Androgen
5.	Ovary	e.	Growth hormone
6.	Testis	f.	Adrenaline

Explanation: The correct match is:
✓ Thyroid gland secretes **thyroxin**
✓ Adrenal gland secretes **adrenaline**
✓ Pancreas secretes **insulin** and **glucagon**.
✓ Pituitary gland secretes **growth hormones, TSH, FSH, LH, ADH** hormones.
✓ Ovary secretes **progesterone** and **estrogen**.
✓ Testis secretes **testosterone (androgen)**.

Post Test – 8

I. Choose the best answer from the given alternatives.

1. Endocrine glands differ from exocrine glands because:
A. They have duct
B. They produce enzymes
C. They are ductless
D. They are nerves

2. The correct match of endocrine glands and the hormones they produce is shown in:
A. Pancreas – Insulin
B. Thyroid gland – Parathyroid
C. Adrenal gland – glucagon
D. Pituitary gland – Thyroxin

3. Which of the following is both an exocrine and endocrine gland?
   A. Pituitary gland C. Adrenal gland
   B. Pancreas D. Thyroid gland
4. The hormone that converts excess blood glucose level into glycogen is:
   A. Insulin C. Thyroxin
   B. Glucagon D. Adrenaline
5. The target organ for insulin and glucagon is:
   A. Kidney B. Brain C. Liver D. Testis

II. **Match the following endocrine glands with their locations.**

Column 'A'	Column 'B'
1. Thyroid gland	a. Top of the kidney
2. Pituitary gland	b. Below the stomach
3. Adrenal gland	c. Male reproductive organ
4. Pancreas	d. Around the neck
5. Testis	e. Base of the brain

2. Thyroid gland

Thyroid gland is located on the front side of the neck region. It produces a hormone called **thyroxin**. A thyroxin hormone consists of mainly proteins and iodine.
The production of thyroxin hormone is stimulated by a pituitary gland by a hormone called **thyroid stimulating hormone (TSH)**.

- ✓ When the concentration of thyroxin in the blood rises above normal, pituitary secretes less TSH.
- ✓ When the concentration of thyroid hormones decrease, the pituitary secretes more TSH.

Thyroid stimulating hormone (TSH) promotes synthesis and secretion of thyroxin.

Thyroxin is used in regulating:
- ✓ metabolic rate of the body
- ✓ utilization of oxygen in the body tissue
- ✓ development of the brain of a growing children.

The production of thyroxin affects the body in different ways:

- ✓ Production of too much thyroxin may cause **thinness** and **restlessness**. This results in an increased metabolic rate which causes loss of weight.
- ✓ Production of too little thyroxin:
  - Makes people feel tired and lack energy.
  - Causes over weightiness.
  - Miscarriage or still birth in pregnant women.
  - In children it causes stunted mental development and physical growth. This condition is called **cretinism**.

The deficiency of iodine in the diet causes swollen or enlarged thyroid gland called **goiter**. This results when the thyroid gland enlarges and swells in search for iodine when it is deficient.

Goiter is a common problem in different regions of Ethiopia. It can be reduced by eating iodized food stuffs such as iodized salt.

3) Parathyroid gland

Parathyroid gland is a butterfly shaped gland located at the back of thyroid gland. It produces a hormone called **parathyroid hormone**. Parathyroid hormone is responsible for regulating the level of **calcium** and **phosphorous** in the blood.

4) Adrenal gland

Adrenal glands are located at the top of each kidney.

- ✓ Its outer region is called **cortex** and the inner region called **medulla**.

Adrenal medulla produces a hormone called **adrenaline**. Adrenaline is also known as a **fight – flight hormone**. It prepares the body for stressful, angry, excited or frightened condition. The adrenaline brings the following changes on the body:

- ✓ Heart beat rate increases
- ✓ Rate of breathing increases
- ✓ Pupil dilates
- ✓ Blood sugar level increase, etc.

v) Pituitary gland

Pituitary gland is a pea sized gland located at the base of the brain. It is responsible for:

- ✓ Secreting several hormones
- ✓ Controlling the activity of other endocrine glands

Pituitary gland is also known as a **master gland**, because it controls the activity of other endocrine glands.

The hormones that are produced by pituitary gland are:

- ✓ **Thyroid stimulating hormone (TSH)** – stimulates thyroid gland to produce the hormone thyroxin.
- ✓ **Growth hormone** – promotes the growth of bones of skeleton and the whole body until the age of maturity.
  - o **Gigantism**– an excess production of growth hormones that result in extra ordinary growth in height.
  - o **Dwarfism** – is a condition which results in shorter height below the normal condition. This is related to too little production of growth hormones.
- ✓ **Follicle stimulating hormone (FSH)** – stimulates the development of follicle in the ovary, maturation and ripening of egg.
  - o It also stimulates the production of estrogen and sperm in men.
- ✓ **Luteinizing Hormone (LH)** – stimulates egg release (ovulation) and testosterone production in males.
- ✓ **Antiduretic hormone (ADH)** – controls water content of the body by its effect on kidneys.

Illustrative Examples

88. The gland that enlarges into goiter as a result of iodine deficiency is:

A. Adrenal
B. Parathyroid
C. Pancreas
D. Thyroid

Explanation:
Thyroid gland which is located around the neck region. The deficiency of iodine results in to an enlargement of this gland. This conditionis called**goiter**.

Answer: D

89. Which of the following gland is not correctly matched with the hormone it produces?

A. Gonads – gonadotrophic
B. Thyroid – thyroxin
C. Pancreas – insulin
D. Pituitary – progesterone

Explanation:
The correct match is:

Gland	Hormone
Gonads	Estrogen and progesterone in female
	Testosterone in male
Thyroid	Thyroxin
Pancreas	Insulin and glucagon
Pituitary	LH, FSH, TSH, growth hormones, ADH

Answer: D

90. Which of the following hormones is secreted by pituitary gland?
A. Leuterizing hormone
B. Parathyroid hormone
C. Progesterone
D. Thyroxin

Explanation: Pituitary gland is a master gland and it secretes the following hormones:
- ✓ Luteinizing hormone (LH)
- ✓ Follicle stimulating hormone (FSH)
- ✓ Antiduretic hormone (ADH)
- ✓ Thyroid stimulating hormone (TSH) and growth hormones.

Answer: A

91. The hormone whose deficiency results in slow growth and stunted mental development in children is:
A. Insulin
B. Parathyroid
C. adrenaline
D. Thyroxin

Explanation: The deficiency of thyroxin in children is called **cretinism**. It causes stunted mental development in children.

Answer: D

92. The endocrine gland which when surgically removed may result in disturbance of blood calcium level is:
A. Iselets of Langerhan's
B. Parathyroid gland
C. Pituitary gland
D. Thyroid gland

Explanation: Parathyroid gland produces a hormone called parathyroid hormone which regulates the level of calcium the blood.

Answer: B

93. What causes goiter?
A. Excessive protein intake
B. Excessive salt intake
C. Iodine deficiency
D. Vitamin deficiency

Explanation: Goiter is a swelling of thyroid gland in the neck region. It is caused by the deficiency of iodine.

Answer: C

467

94. Which of the following glands produces luteinizing hormone?
A. Adrenal C. Pituitary
B. Pancreas D. Thyroid

Explanation: Pituitary gland is a master gland which produces several hormones such as:
- ✓ Growth hormones
- ✓ Luteinizing hormones
- ✓ Follicle stimulating hormone
- ✓ Thyroid stimulating hormone

Answer: C

95. Which of the following hormone causes ovulation?
A. FSH B. Oxytocine C. LH D. Progesterone

Explanation: Luteinizing hormone stimulates egg release (ovulation) and testerone production in males. **Answer: C**

96. Why is the 'pituitary gland' called a master gland?

Explanation: Pituitary gland is called a master gland because:
- ✓ It controls the activity of other endocrine glands.
- ✓ It produces several hormones such as FSH, LH, TSH, growth hormones, etc.

Post Test – 9

I. Multiple choice

1. Enlargement of the thyroid gland is called
A. Thyroidectomy C. Thyrotoxicosis
B. Goiter D. None

2. Hypothyroidism causes in children
A. Myxedema C. Cretinism
B. Goiter D. Diabetes mellitus

3. The pituitary hormone called _____ stimulates the male testes to produce sperm and stimulates the development of the follicle in the female on a monthly cycle.
A. FSH B. LH C. ADH D. TSH

4. Which of the following is not produced by the pituitary gland?
A. Growth hormone C. Luteinizing hormone
B. Thyroxin D. Antidiuretic hormone

5. All of the following are endocrine glands, EXCEPT the:
A. Sweat gland C. Thyroid gland
B. Adrenal glands D. Pituitary gland

6. The endocrine gland (s) that sits on top of the kidneys and secretes both metabolic stabilizers and stress regulators:
   A. Pancreas
   B. Adrenal glands
   C. Thyroid gland
   D. Pituitary gland
7. The endocrine gland (s) referred to as the "master gland" is the:
   A. Pancreas
   B. Adrenal glands
   C. Thyroid gland
   D. Pituitary gland
8. The pituitary gland lies in the:
   A. Abdominal cavity
   B. Inferior aspect of the brain
   C. Along the femoral artery
   D. In the tissues of the neck
9. Name the gland that is located at the base of the throat,
   A. Pineal gland
   B. Thyroid
   C. Hypothalamus
   D. Pituitary
10. Which gland controls basal metabolic rate (BMR)?
   A. Thyroid
   B. Tastes
   C. Parathyroid
   D. Pancreas
11. Which of the following hormones are responsible for the "flight-or-flight" response?
   A. Adrenaline
   B. Insulin
   C. Glucagon
   D. Thyroxin
12. The pituitary hormone that stimulates the male testes to produce sperm and stimulates the development of the follicle in the female on a monthly cycle is:
   A. Growth hormone
   B. Prolactin
   C. Luteinizing hormone
   D. Follicle-stimulating hormone
13. Calcium level in the blood is regulated by the:
   A. Thyroid
   B. Parathyroid
   C. Pancreas
   D. Adrenal medulla

iv) The Gonads

Gonads are glands that produce sex hormones.

- ✓ The male gonad is called **testis**.
- ✓ The female gonad is called **ovary**.

The role of testis

When the boys reach the age of **poverty**, the first stimulates the testis to produce a male hormone called **testosterone**. The testosterone hormone promotes the development of **secondary sexual characteristics**.
The secondary sexual characteristics in male are:

- Public hair, body hair and facial hair begin to grow.
- Growth in height
- Deeping of voice due to enlargement of larynx
- Shoulder and chest broaden
- Enlargement of penis and darkening of skin on the penis.
- Production of sperm by the testis.
- Feeling of independence and questioning develops.

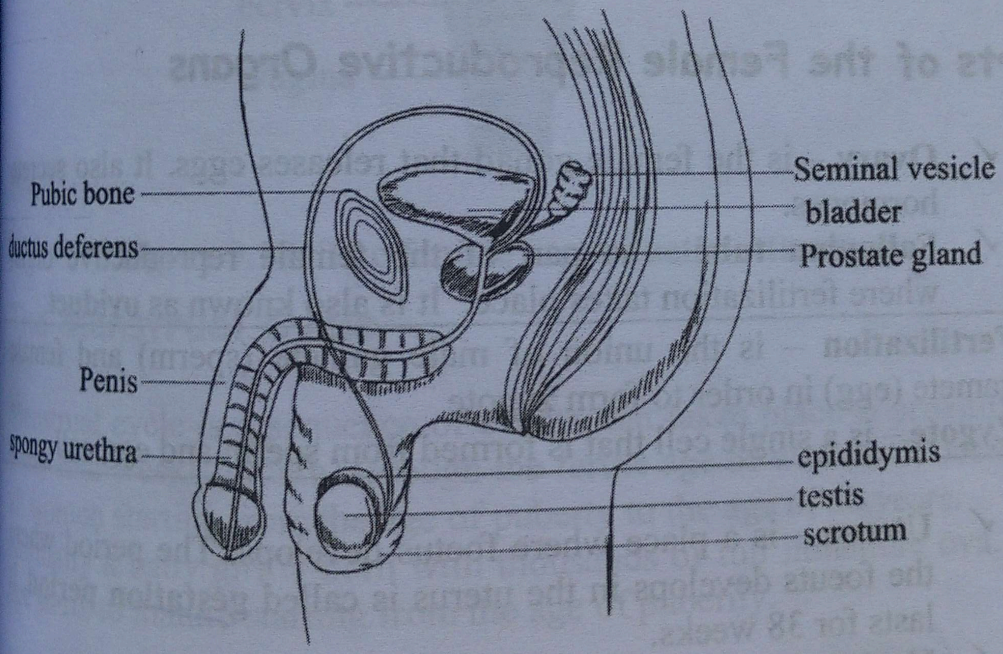


Figure 9.17: The Male reproductive Structure

Parts of Male Reproductive Organ

The male reproductive organ has the following parts:

- ✓ **Testis** – the male gonad that produces sperm and testosterone hormone.
- ✓ **Epididymis**– is the coiled tube where sperm is stored temporarily.
- ✓ **Cowper's gland, prostate gland** and **Seminal vesicles** – produce a fluid called**seminal fluid**.

- ✓ Lubricate sperm cells
- ✓ Transport sperm cells
- ✓ Nourish sperm cells
- ✓ **Urethra** – is a tube through which semen (seminal fluid and sperm cells) are ejaculated.

Ejaculation is the release of semen from the body.

The Role of Ovaries

Ovary is the female gonad. Ovaries are walnut sized organs located in the abdomen. They are associated with fallopian tube (oviduct) and uterus. Ovaries produce two set of hormones that facilitate the female reproductive organ. These are **estrogen** and **progesterone**.

Parts of the Female Reproductive Organs

- ✓ **Ovary** – is the female gonad that releases eggs. It also secrete hormones.
- ✓ **Fallopian tube** – is part of the female reproductive organ where fertilization takes place. It is also known as **oviduct**.

Fertilization – is the union of male gamete (sperm) and female gamete (egg) in order to form zygote.
Zygote – is a single cell that is formed from sperm and egg cell.

- ✓ **Uterus** – is a place where foetus develops. The period where the foeuts develops in the uterus is called **gestation period**. It lasts for 38 weeks.
- ✓ **Vagina** – the part where the sexual intercourse takesplace.
- ✓ **Cervix** – is a part between vagina and uterus. It is known as the neck of uterus.

Secondary Sexual Characteristics in Girls

When a girl reaches the age of puberty, the **FSH** stimulates the ovary to produce sex hormones called **estrogen**. Estrogen stimulates the development of secondary sexual characteristics in girls.
The female secondary sexual characteristics are:

- ✓ Development of height and the whole body structure.
- ✓ Public hair and under the armpits

- ✓ Development of breast
- ✓ Widening of hips
- ✓ The voice becomes thin
- ✓ Maturation of egg and menstruation begins
- ✓ Feeling of independence develops

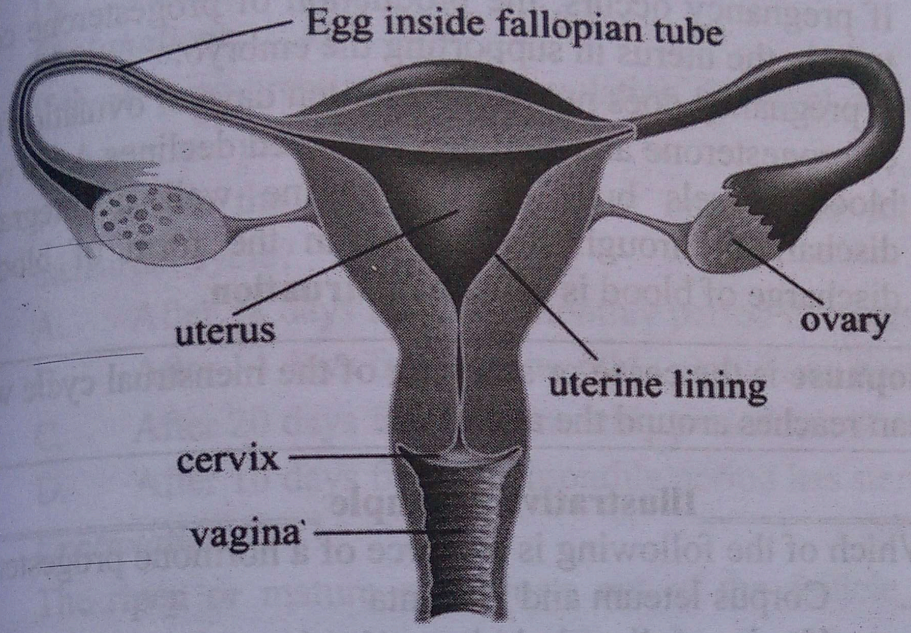


Figure 9.18: The female reproductive structure

The Menstrual Cycle

Menstrual cycle is a sequence of events that takes place approximately every four weeks (month) through the fertile age of women. It occurs in all women starting from the age of puberty to the age of 50 years. Normally a baby girl is born with thousands of full immature ova. The age starts to mature starting from the age of puberity.

Menstrual cycle occurs through the following steps:

- ✓ FSH (follicle stimulating hormone) stimulates the maturation of egg and the ovary to produce a hormone estrogen, every month of the woman's fertile age.
- ✓ A matured egg is released from ovary to fallopian tube always 14 days after menstruation. The process is called **ovulation**.
- ✓ At the same time estrogen starts to stimulate the uterus to build up a thick spongy lining with lots of blood vessels ready to support pregnancy.

- ✓ After ovulation the remains of graffian follicle in the ovary forms **corpus leutum**.
- ✓ The corpus leutum secretes a hormone **progesterone** which keeps the uterine lining thick and growth of more blood vessels on the uterus.
- ✓ If pregnancy occurs, the production of progesterone continues to help the uterus in supporting the embryo.
- ✓ If pregnancy does not occur after ten days of ovulation the level of progesterone and estrogen produced declines.A s a result the blood vessels built on the uterine wall disintegrates and discharged through the vagina in the form of blood. This discharge of blood is called**menstruation**.

Menopause is the cease or stopping of the menstrual cycle when the woman reaches around the age of 50.

Illustrative Example

97. Which of the following is a source of a hormone progesterone?
    A. Corpus leteum and placenta
    B. Uterine wall and pituitary gland
    C. The two fallopian tubes
    D. Uterine wall and young follicles

    Explanation:
    Hormone progesterone is produced by corpus leutum and placenta. It prepares a woman for pregnancy.

    Answer: A

98. Which one of the following produces progesterone?
    A. Oviduct
    B. Ovary
    C. Pituitary gland
    D. Adrenal gland

    Explanation:
    Ovary is a female gonad that produces hormones estrogen and progesterone.

    Answer: B

99. Suppose a girl look like a boy in her external feature, this situation could be due to the lack of the hormone called?
    A. Adrenaline
    B. Testosterone
    C. Insulin
    D. Estrogen

    Explanation:
    Estrogen is a female hormone that gives her a female features and behaviors.

    Answer: D

100. Where there is no pregnancy after ovulation:
A. Follicle is formed in the ovary
B. Estrogen level increases
C. Progesterone level decreases
D. The egg is released from the ovary

Explanation:
If there is no pregnancy after ovulation the level of estrogen and progesterone in the blood declines. **Answer: C**

101. When does the ripening egg brust out of the follicle at the menstrual cycle?
A. After 38 days from the monthly period has started.
B. After 14 days from the monthly period has started.
C. After 20 days from the monthly period has started.
D. After 10 days from the monthly period has started.

Explanation:
The ripen or mature egg brusts out of the follicle (ovary) and moves to fallopian tube 14 days from the monthly menstrual flow has started. This is called**ovulation.**

Answer: B

102. Which of the following is a secondary sexual characteristics of males?
A. Beginning of menstruation
B. Development of breast
C. Broadening of shoulder and chest
D. Production of ova from ovaries

Explanation:
The secondary sexual characteristics of males are:
- ✓ Broadening of shoulder and chest
- ✓ Deeping of voice
- ✓ Appearance of heart under armpits, chest, reproductive organ, face, etc.
- ✓ Ejaculation of sperm cell starts
- ✓ Sexual feeling on opposite sex, etc

Answer: C

The production of thyroxin affects the body in different ways:

- ✓ Production of too much thyroxin may cause **thinness** and **restlessness**. This results in an increased metabolic rate which causes loss of weight.
- ✓ Production of too little thyroxin:
  - Makes people feel tired and lack energy.
  - Causes over weightiness.
  - Miscarriage or still birth in pregnant women.
  - In children it causes stunted mental development and physical growth. This condition is called **cretinism**.

The deficiency of iodine in the diet causes swollen or enlarged thyroid gland called **goiter**. This results when the thyroid gland enlarges and swells in search for iodine when it is deficient.

Goiter is a common problem in different regions of Ethiopia. It can be reduced by eating iodized food stuffs such as iodized salt.

3) Parathyroid gland

Parathyroid gland is a butterfly shaped gland located at the back of thyroid gland. It produces a hormone called **parathyroid hormone**. Parathyroid hormone is responsible for regulating the level of **calcium** and **phosphorous** in the blood.

4) Adrenal gland

Adrenal glands are located at the top of each kidney.

- ✓ Its outer region is called **cortex** and the inner region called **medulla**.

Adrenal medulla produces a hormone called **adrenaline**. Adrenaline is also known as a **fight – flight hormone**. It prepares the body for stressful, angry, excited or frightened condition. The adrenaline brings the following changes on the body:

- ✓ Heart beat rate increases
- ✓ Rate of breathing increases
- ✓ Pupil dilates
- ✓ Blood sugar level increase, etc.

v) Pituitary gland

Pituitary gland is a pea sized gland located at the base of the brain. It is responsible for:

- ✓ Secreting several hormones
- ✓ Controlling the activity of other endocrine glands

Pituitary gland is also known as a **master gland**, because it controls the activity of other endocrine glands.

The hormones that are produced by pituitary gland are:

- ✓ **Thyroid stimulating hormone (TSH)** – stimulates thyroid gland to produce the hormone thyroxin.
- ✓ **Growth hormone** – promotes the growth of bones of skeleton and the whole body until the age of maturity.
  - o **Gigantism**– an excess production of growth hormones that result in extra ordinary growth in height.
  - o **Dwarfism** – is a condition which results in shorter height below the normal condition. This is related to too little production of growth hormones.
- ✓ **Follicle stimulating hormone (FSH)** – stimulates the development of follicle in the ovary, maturation and ripening of egg.
  - o It also stimulates the production of estrogen and sperm in men.
- ✓ **Luteinizing Hormone (LH)** – stimulates egg release (ovulation) and testosterone production in males.
- ✓ **Antiduretic hormone (ADH)** – controls water content of the body by its effect on kidneys.

Illustrative Examples

88. The gland that enlarges into goiter as a result of iodine deficiency is:

A. Adrenal
B. Parathyroid
C. Pancreas
D. Thyroid

Explanation:
Thyroid gland which is located around the neck region. The deficiency of iodine results in to an enlargement of this gland. This conditionis called**goiter**.

Answer: D

89. Which of the following gland is not correctly matched with the hormone it produces?

A. Gonads – gonadotrophic
B. Thyroid – thyroxin
C. Pancreas – insulin
D. Pituitary – progesterone

Post Test – 10

Part I: Choose the best answer from the given alternatives.

1. The male and the female gonads are respectively.
   A. Ovary and testis
   B. Fallopian tube and vasdeferens
   C. Testis and oviduct
   D. Testis and ovaries
2. One of the following is NOT the secondary sexual characteristics of male?
   A. Ejaculation of sperm
   B. Deepening of voice
   C. Monthly flow of menstruation
   D. Enlargement of penis
3. Sperm and testosterone are produced by:
   A. Epididymis
   B. Vasdeferens
   C. Testis
   D. Urethra
4. The temporary storage site of sperm is:
   A. Epididymis
   B. Vasdeferens
   C. Testis
   D. Seminiferous tubule
5. The use of seminal fluid is:
   A. To lubricate the sperm cell
   B. To transport the sperm cell
   C. To nourish the sperm cell
   D. All are possible answers
6. The release of sperm through the penis is:
   A. Copulation
   B. Ejaculation
   C. Fertilization
   D. Mensuration
7. The female gonad produces:
   A. LH and estrogen
   B. Progesterone FSH
   C. Progesterone and estrogen
   D. ADH and testosterone
8. The part of the female reductive organ where fertilization takes place is:
   A. Oviduct
   B. Ovary
   C. Uterus
   D. Cervix
9. The union of sperm and egg cells is:
   A. Unity
   B. Fertilization
   C. Copulation
   D. Intercourse

10. The female secondary sexual characteristics involves all except:
   A. Widening of hips
   B. Broadening of chest
   C. Development of breast
   D. Menstrual cycle starts

11. The release of matured egg from ovary to fallopian tube is called:
   A. Ovulation
   B. Implantation
   C. Copulation
   D. Gestation

12. The stopping or cessation of menstrual cycle in woman after reaching the age of 45 and above is:
   A. Puberty
   B. Menopause
   C. Mensuration
   D. Pregnancy

9.3 Reproductive Health

Sexual intercourse is the most important process for sexual reproduction. The erect penis enters in to the vagina and the semen (seminal fluid and sperm cells) released (ejaculated) into the vagina.

Ejaculation is the release of semen during sexual intercourse.

The journey of sperm along the male reproductive structures is traced as:

Testis → Epididymis →Vasdeferens→ Urethra

After emerging the female reproductive organ the sperm cell passes along the following pathway.

Vagina → Cervix → Uterus → Oviduct (fallopian tube)

The ovum (egg) is released from the ovary to fallopian tube by the process called **ovulation**.
Millions of sperm cells ejaculated fertilize the ovum in the fallopian tube (ovum).
After getting fertilized ovum is called**zygote** and zygote develops into a baby.

Contraceptive (Birth Control) Method

Currently, the world population is growing at an alarming rate. If the population growth of the world continues at its current rate, our planet will face a huge problem especially in relation to the availability of resources.

Contraception is the method of controlling unwanted pregnancy. There are different methods of birth control (contraception). They differ from one another in a way they are used and their effectiveness.

The major contraceptive methods used by people can be classified into the following categories. These are:

- ✓ Natural method
- ✓ Physical or barrier method
- ✓ Intrauterine device (IUD)
- ✓ Hormonal methods
- ✓ Sterilization

i) Natural method

The natural birth control (contraceptive) method is carried without using any physical or hormonal methods. These are:

i) **Rhythm or calendar method** – is based on the understanding of the menstrual cycle and prediction of the ovulation time. This method works perfectly for a woman having a regular menstrual cycle.

ii) **Breast feeding**– is because a woman do not ovulate while she is feeding breast. This is due to the reason that the level of progesterone is the blood is still high to regulate the feeding of breast.

2) Physical or barrier method

Physical method prevents mainly the union of sperm and ovum (egg). These are:

a) **Condom** – is a sheath that is used to prevent the sperm from entering into the vagina. It can be a male or a female condom.

The most commonly used type of condom is the male condom. This is due to the reason that the female condom is relatively expensive.

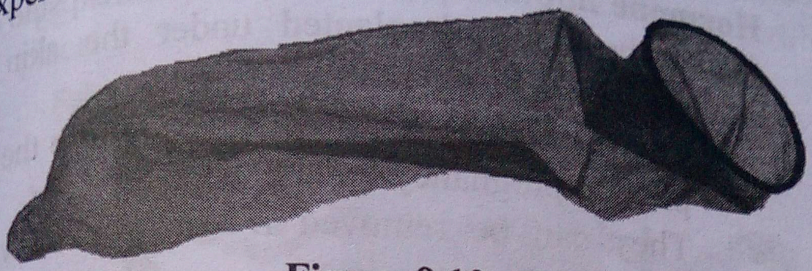


Figure 9.19: Condom

b) **The diaphragm or cap** – is a thin rubber inserted into the vagina before the sexual intercourse. It prevents the entry of sperm on a cervix.

Figure 9.20: Diaphragm

3) Hormonal method

Hormonal method involves the use of natural hormones (estrogen and progesterone) to prevent pregnancy. These are:

i) **Mixed pill** – consists of the hormones estrogen and progesterone. This rise the level of estrogen and progesterone in the blood.
→ Rising of the level of estrogen in the blood is detected by the pituitary gland. This slows down the production of Follicle stimulating hormone (FSH).

FSH is responsible for maturation of egg and the absence of FSH prevents maturation of the egg. If the egg do not mature, there will be no fertilization and pregnancy.

The progesterone increase the mucus in the cervix and vagina to prevent the entrance of sperm cell.

ii) **Hormone implants** – are small silicon capsule containing female hormones implanted under the skin (arm) of women.
- The hormones are released slowly into the blood and prevent pregnancy for five years.
- They can be removed if the women want to get pregnant.

iii) **Hormones injections** – is a method of injecting the hormones estrogen and progesterone. This increases the level of hormones in the blood.
- It can prevent pregnancy up to three months.

4) Sterilization or Surgical Contraception

This method makes people not to reproduce by making them sterile. Sterilization method can be applied to both men and women.

✓ **Vasectomy** – is a male sterilization in which involves cutting and trying vasdefernes (sperm duct).
→ This prevents the passage of sperm along the urethra.
→ Once cut and tied it is difficult to restore. Thus, such a method requires a big decision.

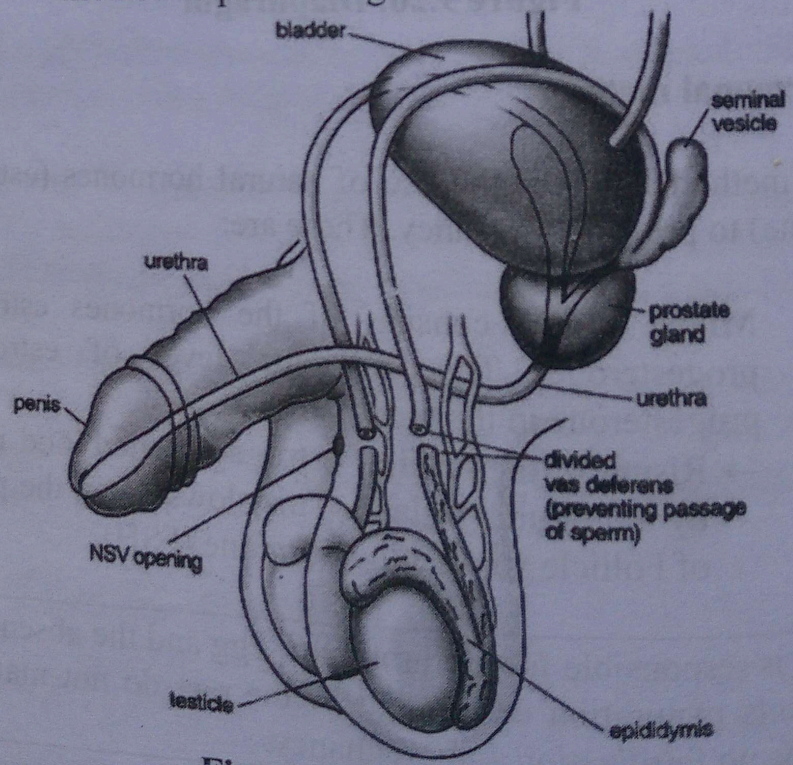


Figure 9.21: Vasectomy

✓ **Tubectomy**– is a female sterilization. It involves cutting and tying the oviduct (fallopian tube). This prevents the release of egg from ovary to fallopian tube where fertilization takes place.

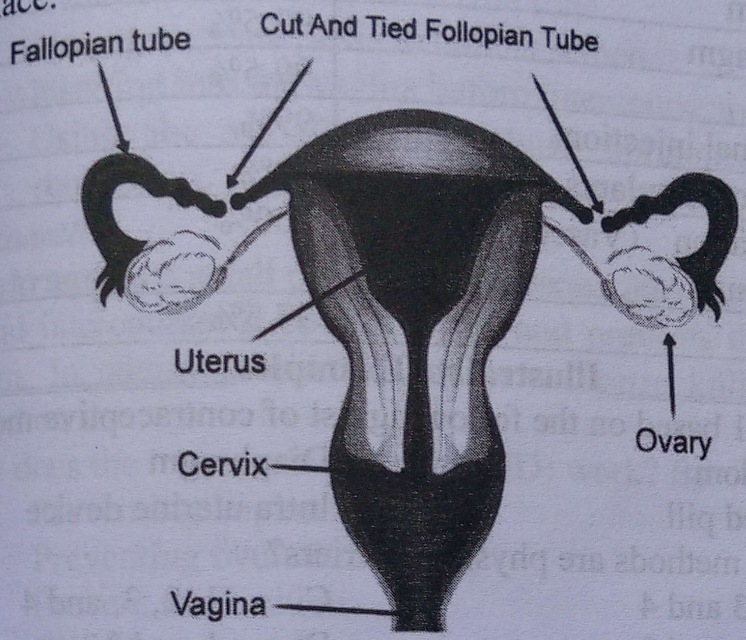


Figure 9.22: Tubectomy

5) Intrauterine device (IUD)

Intrauterine devices are made up of plastic and copper that is inserted into the uterus by a doctor. IUD'S do not prevent fertilization but it prevents implantation.

Implantation – is the process by which the foetusgetsattached on the uterine wall.

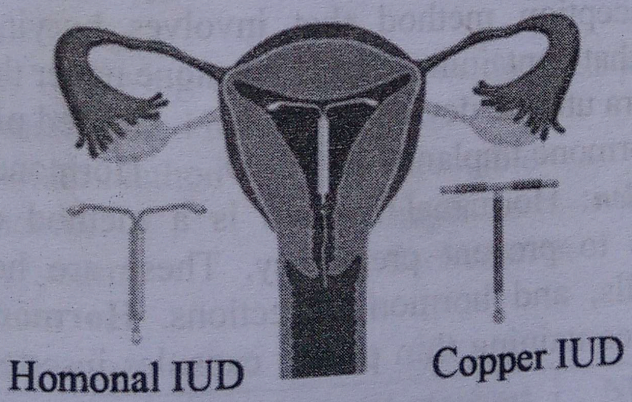


Figure 9.23: IUD

Table: Effectiveness of the Contraceptive methods

Contraceptive method	Effectiveness
Rhythm or calendar method	90%
Condom	97.5%
Diaphragm	97.5%
Pills	99.5%
Hormonal injections	99%
Hormonal implants	99%
Sterilization (Vasectomy and tubectony)	100%
IUD	97.5%

Illustrative Examples

Item number 1 based on the following list of contraceptive methods.

1. Condom
2. Mixed pill
3. Diaphragm
4. Intra uterine device

110. Which methods are physical barriers?

A. 3 and 4
B. 1 and 2
C. 1, 2, 3, and 4
D. 1 and 3

Explanation: Physical barriers are:

- ✓ Condom and
- ✓ Diaphragm

Answer: D

111. Which of the following contraceptive methods prevents the implantation of the early embryo?

A. The mixed pills
B. Sterilization
C. The diaphragm
D. The intra uterine device

Explanation: Intra uterine devices are contraceptive methods that are capable of preventing implantation but not fertilization.

Answer: D

112. A contraception method that involves burying small silicon capsules that containing female hormone under the skin is

A. Intra uterine device
B. Hormone implant
C. Mixed pills
D. Hormone injection

Explanation: Hormanal method is a method of using female hormones to prevent pregnancy. These are hormone implant, mixed pills, and hormone injections. **Hormone implants** are hormone containing thin plastic capsules inserted under the skin of the arm.

Answer: B

113. How does the diaphragm work to control unwanted conception? By
A. Avoiding sex during the fertile time.
B. Placing over the penis to collect semen
C. Inserting into the vagina before intercourse to cover cervix
D. Using the variation of natural hormones to prevent conception.

Explanation:
Diaphragm is a small plast rubber inserted into the vagina before sexual intercourse. It covers cervix and prevents the passage of sperm. In addition, spermicidal foam (sperm killing cream) is putted on it.

Answer: C

114. How does the intrauterine device (IUD) work? By
A. Preventing implantation
B. Preventing ovulation
C. Prevent fertilization
D. Killing sperm cells

Explanation: See illustrative example number 111.

Answer: A

115. List out the following contraceptive methods based on decreasing level of effectiveness.

✓ Condom	✓ Calendar method
✓ Pills	✓ Hormonal implants
✓ Sterilization	

Explanation: According to their level of effectiveness from the most effective to least effective onethey are arranged as:
Sterilization → pills → hormonal implants → condom → calendar method

116. Identify the following contraceptive methods as physical method, hormonal method and natural method.

Calendar method	Injectable
Condom	Implants
Diaphragm	Sterilization
Pills	

Explanation:

Physical method	Hormonal method	Natural method
Condom Diaphragm	Pill Injectable Implants	Rhythan method Breast feeding

Post Test – 11

I. Choose the best option from the given alternatives.

1. The correct path of sperm flowis?
   A. Epididymis→ Testis →Vasdeferens→ Urethra
   B. Testis →Epidiymis→Vasdeferens→ Urethra
   C. Urethra → Testis →Vasdeferens→ Epididymis
   D. Vasdeferens→ Urethra →Epidymis→ Testis
2. Which one of the following is a physical birth control method?
   A. Condom C. IUD
   B. Pill D. Injectable
3. Which of the following is NOT a natural birth control method?
   A. Breast feeding C. Calendar method
   B. Rhythm method D. Norplant
4. Small silicon capsules containing female reproductive hormones and inserted under the skin are:
   A. Injectables C. Vasectomy
   B. Implants D. Condom
5. The sterilization of male is:
   A. Tubectomy C. Vasectomy
   B. Castiration D. Clitoridectomy
6. Rearrange the following contraceptive methods according to increasing level of their effectiveness
   i. Condom iv. Pills
   ii. Sterilization v. Calendar method
   iii. Hormonal implant

9.4 Homeostasis

Organisms must make adjustments, which combat the changes that have occurred in their environment. This applies for both to the changes taking place in the **internal** as well as **external** environments.

The process of maintaining constant internal environment by an organism to keeps the body's function normal and stable is called**homeostasis.**

The word homeostasis is derived from a Greek word '**homios**' and '**stasis**'.

- ✓ **'Homios'** means **'like'** or **'the same'**
- ✓ **'stasis'** means **'state'**

Homeostasis works to keep body's stability for the following metabolic process:

- ✓ Sugar (glucose) level of the blood
- ✓ Body temperature regulation
- ✓ Salt and water balance of the body
- ✓ Removal of excretory products
- ✓ PH level of the body, etc.

The Organs of Homeostasis

Different organs in the body work in keeping or metabolic system stable. These organs are:

- ✓ Skin
- ✓ Kidneys
- ✓ Lung
- ✓ Liver

1. Temperature regulation

Temperature is one of the environmental factors that organisms have to cope with.

No animals can live if its temperature goes beyond certain limits. Thus, to keep (stay) alive an animal must avoid extremes of temperature, either too cold or too hot.

Regulation of body temperature is important for animal to keep various metabolic activities of the body stable. For instance, the proper working of our body enzymes is determined by our body temperature.

Enzymes are biological catalysts in our body. They keep to regulate the speed of the chemical reactions in our body. There are thousands of reactions in our body that are catalyzed by enzymes. Enzymes are sensitive to temperature.

- ✓ Under extremely high temperature enzymes get denatured.
- ✓ At lower temperature than the optimum temperature of the body, enzymes become inactive.

For example enzymes in body are active 37°C which is optimum for the action of enzymes in the body.

Regulation of body temperature is maintained by the heat loss and gain of the body.

- ✓ Organisms gain heat by mechanism such as **respiration, conduction, convection** and **radiation**.
- ✓ Organisms lose heat by mechanism such as **evaporation of water from the body surfaces convection** and **radiation**.

Meanings of Important Terminologies

- ✓ **Radiation** – is the diffusion of heat from the warmer body to a relatively colder objects with the air.
- ✓ **Evaporation** – is evaporation of water from the surface of the body.
- ✓ **Conduction** – is the transfer of heat from the hotter to the cooler object when two objects are in contact.
- ✓ **Convection** – is the transfer of heat between an organism and the environment with the currents of air or water which is in contact with the surface of an organism.

Therefore, the heat gain and heat losmust be balanced based on the body's need.

Piokilotherms and Homeotherms

Based on their temperature regulation mechanisms organisms can be classified as **piokilotherms (ectoderms)** and **homeotherms (endothermic)**.

i) **Poikilotherms** are organisms that have a body temperature whichvaries with change in environmental temperature.

ii) **Homeotherms**– are organisms which have constant body temperature which remains constant despite changing environmental temperature.

Examples: Birds and mammals

Table: Comparison of piokilotherms and homeotherms

Homotherms	Poikilotherms
Have constant body temperature	Have variable body temperature
Need to consume more food	Consume less amount of food
Produce more internal heat (20 times that of poikilotherm)	Produce little internal heat

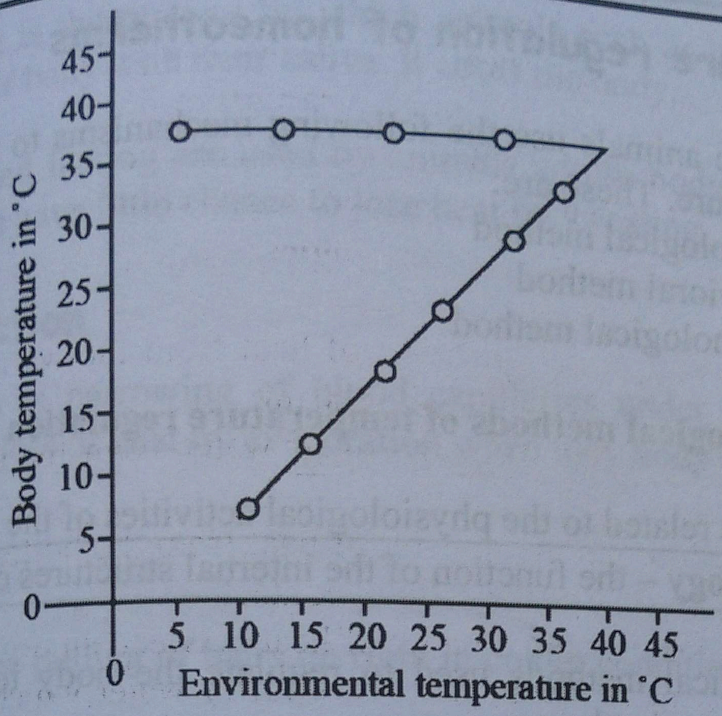


Figure 9.24: Body temperature of a cat and lizard under different temperatures

Temperature regulation in poikilotherms

Poikilotherms use different mechanisms for regulating their body temperature.

In a colder condition they regulate their body temperature by:

- ✓ Basking themselves in the sun.
- ✓ Pressing their body close to warm surface.
- ✓ Erecting special sails or areas of their skin which allows them to absorb more heat.

As we can clearly see from the above graph or figure:

- ✓ Cat is a homoeothermic animal. Thus, it is clearly shown, whatever the environmental temperature is the cats body temperature remained almost constant.
- ✓ Lizard is a poikilothermic animal. Thus, as its shown above the body temperature of lizard inclines (increases) with rising environmental temperature.

When the climate is very hot piokilotherms:

- ✓ Move into the shade
- ✓ Immerse into water bodies or mud.

Temperature regulation of homeotherms

Homeothermic animals use the following mechanisms to regulate their body temperature. These are:

- ✓ Physiological method
- ✓ Behavioral method
- ✓ Morphological method

a) Physiological methods of temperature regulation

This method is related to the physiological activities of the body.

Physiology – the function of the internal structures of the body.

The physiological methods used to regulate the body temperature in homoeothermic animals are:

Sweating

Sweating is the process by which heat evaporates (oozes) from the body in the form of sweat.
Sweat largely consists of water and nitrogenous wastes to a little extent.
It has a cooling effect on the body.

Vasodilatation

Vasodilation is the process in which the capillaries under the skin surface dilate when the body temperature rises.

- ✓ Dilation of capillaries under the skin brings more blood on the skin surface with more heat to be evaporated.
- ✓ Vasodilation has cooling effect on the body.

Panting and Licking

- ✓ **Panting** – allows losing more heat through mouth in the form of evaporation. It is used by mammals such as dog. It has cooling effect on the body.

- ✓ **Licking** is the process by which animals such as cat and dog lick their body with their saliva. It cools the body.

Panting and licking are used by animals whose body is covered by fur and have little chance to lose heat by sweating.

Vasoconstriction

Vasoconstriction is narrowing of blood capillaries under the skin in order to prevent loss of heat by evaporation when they body temperature fails.

Piloerection

Piloerection is the pulling of hairs up right in colder condition. In such a condition:

- ✓ Erector muscles under the skin contract. This pulls the hairs upright to trap an insulating layer of the skin to conserve more heat.

Shivering

Shivering results as a rapid contraction and relaxation of muscles of the body. This results in

- ✓ Increasing the rate of respiration to generate more heat in a condition when a body feels colder.

Fat Layer (Subcutaneous Fat)

Fay layer under the skin prevents excessive loss of heat form the body.

Fat has high heat storage capacity.

Animals living in cold climatic regions have thick layer of fat under the skin to keep the body warm.

b) Behavioral methods of Temperature Regulation

Animals living in different climatic conditions have their own behavioral adaptations by which they react to temperature changes. These are:

- ✓ **Clothing** – humans follow different wearing styles based on the climatic conditions of their environment.
- ✓ **Seeking shade or shelter** – in a warmer condition animals hide or seek in shades or shelter to regulate excessive heat coming from the sun.
- ✓ **Taking high calorie food** – food stuffs containing high calorie such as carbohydrates help the body to generate internal heat. This helps in a colder climate.
- ✓ **Hibernation** – is a condition in which organism go into a very deep sleep in colder seasons. During this condition the whole metabolic rate drops.
- ✓ **Aestivation** – is a condition in which some animals hide underground or under a layer of mud and go in to a deep sleep in warmer conditions.

The body temperature is regulated by a regulatory center in the brain called **hypothalamus.**

- ✓ When the body temperature rises, hypothalamus coordinates the body temperature to become normal by:
  - → making the sweat glands to produce sweat.
  - → making erector muscles of the skin relax and make the hair flat.
  - → arterioles on the skin becomes wide (vasodilation)
  - → tyroid gland produces less thyroxine
- ✓ When the body temperature decreases, hypothalamus coordinates the body temperature to become normal by making:
  - → Sweat glands not to produce sweat.
  - → Erector in the muscles of the skin contract and make the hair raised (erect).
  - → Arterioles in the skin become narrow (Vasoconstriciton)
  - → Thyroid gland produce more thyroxine

c) Surface area to Volume ratio

Surface area to volume ratio (SA/V) represents the surface of the body pin relation to its volume.

Temperature exchange between an organism and its surrounding occur at the skin, is depending of the surface area of the skin.

- ✓ Smaller animals have large surface area to volume ratio. This implies that they lose heat faster to their surroundings.
- ✓ Bigger animals have relatively smaller surface area to volume ratio. This shows that, they do not lose heat faster to their surroundings.

Illustrative Examples

117. Which of the following physiological method of temperature regulation occurs in homoeothermic animals when temperature begins to fall?

A. Vasodilation
B. Panting
C. Licking
D. Vasoconstriction

Explanation: Vasodilation, panting and licking are temperature regulation mechanisms that are used by homoeothermic animals when temperature rises. They result in cooling the body. Vasoconstriction is used to conserve or save internal heat when temperature begins to fall.

Answer: D

118. Which of the following organisms' body temperature is governed by environmental temperature?

A. Lizard B. Sheep C. Ostrich D. Lion

Explanation: Lizard is a poikilthermic animal which body temperature is regulated by environmental temperature.

- Sheep, ostrich and lion are homoeothermic animals that can regulate their internal body temperature.

Answer: A

119. Which of the following is a physiological method of temperature regulation in homeotherm?

A. Hibernation
B. Festivation
C. Bathing
D. Panting

Explanation:
There are different methods of temperature regulation. Accordingly,

- ✓ Hibernation, aestivation and bathing are behavioral methods of temperature regulation.
- ✓ Panting is a physiological method of temperature regulation.

Answer: D

121. This item is based on different processes listed below.

1. evaporation
2. conduction
3. convection
4. cellular respiration
5. radiation

Which of these are heat losing processes?

A. 1, 3, 4, 5
B. 1, 2, 3, 4
C. 2, 3, 4, 5
D. 1, 2, 3, 5

Explanation:

Heat losing process	Heat gaining processes
- Evaporation - Conduction - Convection - Respiration	Radiation

Answer: B

122. An animal whose internal temperature varies along with that of the ambient environmental temperature is known as

A. homeostasis
B. aestivation
C. homeotherms
D. poikilotherms

Explanation: see example number 119. **Answer: D**

123. Which of the following is true about homeostasis?

A. Decreases the level of glucose in our body
B. Raise the level of glucose in our body
C. Maintains the level of Co_2 in our body
D. Increases the level of Co_2 in our body

Explanation: Homeostasis helps to regulate constant internal environment. This may include:

✓ Level of glucose in the body
✓ Level of Co_2 in the body, etc.

124. Which of the following mechanisms is physiological method of temperature regulation?

A. Sweating
B. Aestivation
C. Clothing
D. Bathing

Explanation: Sweating is a physiological method of temperature regulation, while, aestivation, clothing and bathing are behavioural methods of temperature regulation.

Answer: A

125. What is the significance of homeostasis? It is important to

A. Increase the cellular activities of our body systems
B. Denature the enzymes and stop cell activities in our body
C. BUrn food molecules to produce carbon dioxide
D. Maintain the internal conditions of the body in stable state

Explanation: Hemostasis is the process of maintaining constant internal environment or keeping the body stable. **Answer: D**

126. When poikilotherms get too hot they
A. Move into shade
B. Bask in the sun
C. Press their body in warm surface
D. Erect special sail of their skin

Explanation: Poikilotherms
- ✓ When they get too hot, they move in to a shade.
- ✓ When they get to cold they:
  - bask in the sun
  - press their body in to warm surface
  - erect special nails of their skin

Answer: A

127. Which of the following organisms is a poikilothermic animal?
A. Human
B. Ape
C. Elephant
D. Fish

Explanation: Fish is a poikilothermic animal, while human, elephant and ape are homeothermic animals

Answer: D

128. Which of the following methods is correct about regulation of temperature in reptiles?
A. panting
B. Vosoconstriction
C. Bask in the sun
D. Sweating

Explanation: Repitiles such as lizard and corocodile bask themselves in the sun to get heat from their environment.

Answer: C

129. Which of the following statement is correct about poikilotherms?
A. reduce their metabolic rate by shivering
B. release large amount of sweat during hot temperature
C. lose heat through panting during hot temperature
D. Move into water during very hot temperature

Explanation: Shivering, sweating and panting are physiological methods temperature regulation homeothermic animals.
- Poikilothermic animals use mechanisms such as basking in the sun, moving into water bodies to regulate their body temperature.

Answer: D

130. Organisms that have constant internal body temperature are known as:
A. cold blooded
B. environmental dependent
C. homeothermic
D. poikilothermic

Explanation:
- Homeothermic animals have constant internal body temperature that is not affected by external temperature
- Poikilothermic animals are animals that body temperature varies with change in environmental temperature.

Answer: C

131. Which of the following mechanism is a physiological method of temperature regulation?
A. bathing
B. clothing
C. hibernation
D. sweating

Explanation: Sweating is physiological heat loss mechanism, While; bathing, clothing and hibernation are behavioral methods of temperature regulation. **Answer: D**

132. What is homeostasis? It is
A. Absorption of food in digestive systems
B. Maintenance of constant internal environment
C. Adaptation of human beings to their environment
D. Production of red blood cells from the bone marrow

Explanation: See example number 125 **Answer: B**

133. Which one of the following is correct about the adaptation of snakes to high temperature?
A. Sweating
B. Vasoconstriction
C. Basking in the sun
D. Moving into shade

Explanation: Snake is a poikilothermic animal. Thus, it moves in to a shade while exposed to high temperature.

Answer: D

134. How do the poikilotherm adapt to cold weather? By
A. Basking in the sun
B. Bathing in water
C. Licking their body parts
D. Wallowing mud

Explanation: Piokilotherms adapt to cold weather by basking themselves in the sun. **Answer: A**

135. Dialation of the blood vessels supplying blood to the capillaric in the skin is the skin is a response to
A. low oxygen concentration
B. hot temperature
C. cold temperature
D. high acidity

Explanation: Vasodialation is a dialation of blood capillaries under the skin in response to hot temperature. **Answer: B**

136. Identify the following methods of temperature regulation as physiological or behavioral method.

- ✓ vasodialation P
- ✓ sweating P
- ✓ bathing B
- ✓ hibernation B
- ✓ vasoconstriction P
- ✓ aestivating B
- ✓ shivering P
- ✓ clothing B
- ✓ panting P
- ✓ piloerection P

Explanation:

Physiological methods	Behavioral methods
- Vasodilatation - Sweating - Vasoconstriction - Shivering - Panting - Piloerection	- Bathing - Hibernation - Aestivation - Clothing

137. Categorize the following methods of temperature regulation as heat loss or heat gain methods.

- Piloerection G
- Aestivation L
- Sweating L
- vasoconstriction G
- Shivering G
- Vasodilatation L
- bathing L
- Hibernation L
- panting L

Explanation:

Methods of heat loss	Methods of heat of gain
- Sweating - Vasodilatation - Bathing - Hibernation - Panting	- Piloerection - Aestivation - Vasoconstriction - Shivering

Post Test – 12

Part I: Choose the best option from the given alternatives.

1. Which of the following animal has variable body temperature?
   A. Cat B. Dog C. Zebra D. Lizard
2. Among the following temperature regulating mechanisms, which ones are mechanisms of heat loss?
   A. Vasoconstriction
   B. Shivering
   C. Bathing
   D. Licking

3. The largest portion of the sweat is:
   A. Salt B. Water C. Heat D. Urea
4. Homoeothermic animals differ from poikilotherms because they:
   A. depend on external heat
   B. have variable body temperature
   C. are cold blooded
   D. have constant body temperature
5. Which of the following phrases best describes homeostasis:
   A. the ability of the body to constantly change its internal environment
   B. the body's natural tendency to change its internal temperature based on its external temperature
   C. the relatively constant internal physical and chemical conditions that organisms maintain despite the external environment
   D. A and B
6. All of the following belong to the same process except:
   A. Vasodilatation C. Panting
   B. Sweating D. Shivering
7. Define the term homeostasis in relation to temperature:
   A. Maintenance of processes to increase the body temperature
   B. Maintenance of processes to keep a constant body temperature
   C. Maintenance of processes to keep the skin cool
   D. None of the above
8. Identify the method through which heat is not lost from the body:
   A. Sweating
   B. Panting
   C. Vasoconstriction
   D. Vasodilation
9. Identify the term that describes what happens to blood vessels when the condition is hot:
   A. Vasoconstriction C. Sweating
   B. Vasodilatation D. Shivering
10. Identify the correct piece of equipment for taking an animal's temperature:
   A. Stethoscope C. Thermometer
   B. Sphygmomanometer D. None of the above

11. The procedure of maintaining the amounts of salt and water in tissue fluids and blood is called
A. homeostasis
B. excretion
C. osmoregulation
D. thermoregulation

12. The procedure of maintaining the internal temperature of the body is classified as
A. Thermoregulation
B. Osmoregulation
C. Excretion
D. Glucoregulation

13. If the change in temperature occurs then this change may affect
A. Fasten up the metabolism
B. Functions of enzymes
C. Process of diffusion
D. Process of osmosis

14. The process in which metabolic wastes are to be eliminated from the body to maintain the balance of internal condition is called
A. thermoregulation
B. osmoregulation
C. binary fission
D. excretion

15. Which of the following thermoregulatory responses would be considered a strictly behavioral response (not physiological)?
A. Shivering
B. Hibernation
C. Sweating
D. Basking in the sun

16. As the size of an animal increases:
A. The surface area to volume ratio increases
B. The rate of heat loss from the body increases.
C. The amount of heat intake per day decreases.
D. A and C
E. None of the above

17. Poikilothermic animals are those animals:
A. Whose blood is always cold
B. Whose blood is always warm
C. Whose body temperature varies
D. Whose body temperature is constant

18. Which of the following is a method of reducing heat loss from the body?
A. Panting
B. Vasoconstriction
C. Vasectomy
D. Vasodilation

19. Which of the following is NOT important for body's heat conservation?
    A. Subcutaneous fat
    B. Feather or fur cover
    C. Vasoconstriction
    D. Large surface area to volume ratio

2. Excretion

Excretion is the removal of:
- Waste substances produced by metabolism from the body
- Substances which are excess in the body

Metabolic wastes or excretory substance are removed through organs of the body known as **excretory organs**.

The major excretory organs of the body are:
- ✓ **Lungs** are organs of gas exchange and are also excretory organs. They are involved in the removal of carbon dioxide which is produced as a byproduct of respiration.
- ✓ **Skin** is an excretory organ which is involved in removal of excess substances such as **salt**, **water** and **nitrogenous wastes** from the body.
- ✓ **Kidney** is an excretory organ that removes metabolic wastes such as urea, excess water and excess salt in the form of urine.

The processes of excretion and egestion are different from one another.
- ✓ **Excretion** is the removal of excess substances and metabolic wastes from the body.
- ✓ **Egestion** is the removal of indigestible substances from the body, through the anus.

The major excretory organs from the body are:
- ✓ **Carbondioxide**– produced as a result of cellular respiration and the breakdown of urea.
- ✓ **Water** – produced by aerobic respiration.
- ✓ **Urea (nitrogenous waste)** produced by the liver during the process of deamination.

Deamination is the process in which excess protein (amino acid) removed from the body.
It involves removal of the amino group (NH_2) which changes in to urea. This process occurs in the liver.

3. Water Balance and Ionic control

Osmoregulation is the process of regulating water and ionic (electrolyte) balance of the body. The amount of daily water gained and lost must be in balance.

Table: The average daily water intake and output in adult human per day

Water intake		Water loss	
Source	Amount in cm^3	Source	Amount in Cm^3
Drinking	1400	Urine	1500
Food	800	Sweat	600
Cell respiration	400	Breathing (exhalation)	400
		Feaces	100
Total	**2600cm^3**	**Total**	**2600cm^3**

As shown in above table the amount of water that an adult human being takes in and removes from the body is in balance.

Balancing the ionic concentration of the body is vital.

- ✓ Mineral ions are taken in to the body with the foods we eat.
- ✓ Ions are removed from the body as excretory products in the form of sweat and urine.

The kidney

Kidneys are bean shaped reddish brown organs located in either sides of the waist.

Kidneys are both **excretory** and **osmoregulatory** organs of the body. They filter blood and remove excess water, excess salt and nitrogenous wastes such as urea in the form of urine from the body.

Longitudinal section of the kidney

The outer region of the kidney is called **cortex**, while, the inner region of the kidney is called **medulla**. The **medulla** consists of **pyramids** that open in the a hallow chamber called **pelvis**.

Each kidney consists of millions of nephrons in which **blood filtration** and **real sorption** takes place.

Nephrons are the functional units of the kidney.

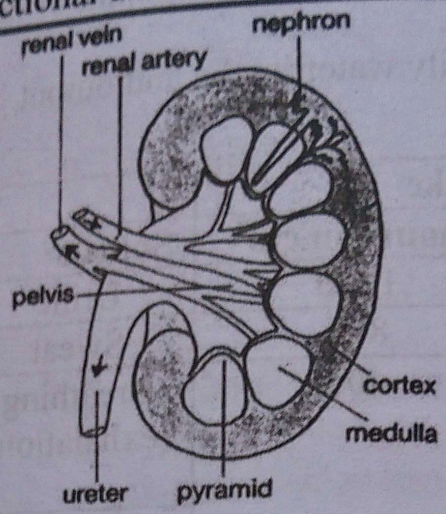
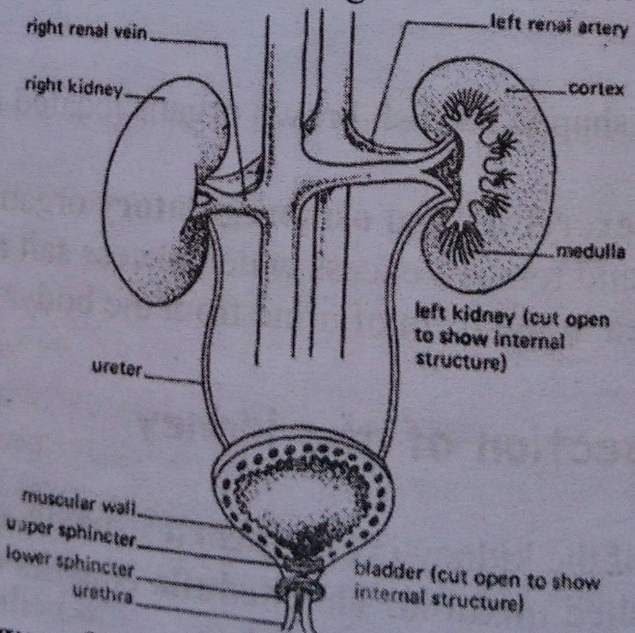


Figure 9.25 Longitudinal sections of the kidney

The Human urinary system

Human urinary system consists of the organs of the body which are involved

- ✓ **Ureter** – a tube which carries urine from the kidney to the urinary bladder.
- ✓ **Urinary bladder** – is a temporary storage site of urine.
- ✓ **Sphincter muscles** – are special muscles at the base of the bladder to control the release of urine in to urethra.
- ✓ **Urethra** – is a tube that discharges the urine out of the body.



igure 9.26: The human urinary system

- ✓ **Renal artery** – supplies oxygenated blood to the kidney.
- ✓ **Renal vein** – carries deoxygenated blood away from the kidney in to the inferior venacava.
- ✓ **Collecting duct** – transports urine down through the medulla and deliver it in to the pelvis.
- ✓ **Afferent arteriole** – supplies blood to glomerulus which are found in the Bowman's capsule
- ✓ **Efferent arteriole** takes blood from glomerulus to the renal tubules.

Afferent arterioles are much wider than **efferentarterioles**. This creates a pressure in glomerulus as the blood moves from afferent arteriole to efferent arteriole.

The urine is formed in two main stages. These are:

- ✓ **Ultra (pressure) filtration**
- ✓ **Reabsorption**

Stage 1: Ultra (pressure) filtration

This stage is concerned with separation of larger substances from smaller ones by a very fine filter as a result of the pressure created in the glomerulus. Ultra filtration takes places through the following sequential orders:

i. Blood enters glomerulus at high pressure.
ii. As a result substances with smaller size like water, salt, glucose, urea and others are forced out of glomerulus to renal tubule. These substances are called **glomerular filtrate**.
iii. Substances that have large size such as blood cells and blood proteins cannot leave glomerulus and are passed to efferent arteriole.

Quick check

Identify substances that are filtrated by glomerulus.

- Blood proteins
- Water
- Salt
- Urea
- Blood cells
- glucose

Stage 2: Reabsorption

Reabsorption is the stage in which useful substances need by the body are reabsorbed back into the body from the glumerullar filtrate.